

B.E.A.C.O.N.



Bicycle Electronic Awareness and Communication On-road Network

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1.0 Executive Summary

As bicycle transportation continues to increase in popularity, cyclist safety has become an increasingly important concern. Unlike motor vehicles, bicycles generally lack integrated systems for communicating braking, turning, and other rider intentions to surrounding traffic. Many cyclists rely on hand signals and passive reflectors, which can be difficult for motorists to see and may require riders to remove a hand from the handlebars. These limitations reduce communication between cyclists and drivers and contribute to the risk of roadway accidents.

To tackle this issue, we have developed a solution called the Bicycle Electronic Awareness and Communication On-road Network, or B.E.A.C.O.N for short. This system is designed to make biking safer by improving rider visibility, communication, and awareness of the rider's surroundings. B.E.A.C.O.N combines several key features, including automatic brake lights, electronic turn signals, and radar that detects when vehicles are nearby. These features work together to alert riders to potential hazards. By putting all these features into one integrated platform, B.E.A.C.O.N aims to reduce the risks that cyclists face on the road and make rides safer and less stressful.

The system is made up of two main parts. One part is an electronics box that fits under the bicycle seat. It contains the microcontroller, radar sensor, inertial measurement unit, battery, and lights that show when the rider is braking or turning. The other part is a control box mounted on the handlebars. This box allows the rider to activate the turn signals and interact with the system. When these two parts work together, they help the rider communicate with nearby cars and trucks, while also helping the rider know when a vehicle is approaching from behind.

The main aim of this project is to make cycling safer by helping cyclists be seen and understood more clearly. It does this in a few ways. First, it automatically turns on the brake light when the cyclist is slowing down, so nearby drivers know what is happening. It also has electronic turn signals that clearly show which direction the cyclist is going. In addition, it has a radar system that detects when vehicles are coming up behind the cyclist and warns the rider. By putting all these features together, B.E.A.C.O.N hopes to reduce confusion between cyclists and drivers. The system is also designed to be affordable, compact, and compatible with regular bicycles. This allows cyclists to feel safer on the road while helping drivers better understand what cyclists are doing.

This paper explains the process of designing the B.E.A.C.O.N system. First, we discuss the motivation and background behind the project. Then, we examine the research conducted on the technologies used to build the system. This includes how the different components were selected, how the hardware and software work together, how the power system was designed, what communication

protocols were used, and how the subsystems were integrated. The paper also discusses how the system was tested and what was learned from managing the project. Finally, the paper summarizes the findings and includes additional supporting information in the appendices.

2.0 Project Description

2.1 Motivation and Background

In recent years, bicycle transportation has experienced significant growth. This development follows the culmination of several compounding shifts in social perception, including environmental awareness, urban congestion, rising fuel costs, and an increase in personal fitness following the recovery of the COVID-19 pandemic. As the popularity of cycling has increased, the industry surrounding it has responded similarly. Changes in new legislation, city planning, and notably the popularity of E-bikes have helped to make biking more accessible, increasing the number of riders on the street [SSS1]. Despite these trends, improvements in biker safety have yet to match, resulting in a rise in related accidents. As more cyclists start to share roadways with motor vehicles, rider safety continues to become an increasingly important engineering concern [SSS2] [SSS3].

Of these concerns, one of the largest contributors to bicycle-related accidents is reduced rider visibility and limited communication, which is compounded during nighttime riding or in dense traffic conditions [SSS4]. This is because, unlike traditional roadway vehicles, bicycles lack integrated signaling systems to communicate intentions such as braking, turning, or lane changes. Many cyclists rely solely on reflectors or hand gestures, the latter of which requires releasing the handlebars, reducing vehicle control. As a result, there is a growing demand for compact, low-power, intelligent safety systems that improve rider visibility while remaining lightweight and affordable [SSS5].

Within the industry, many modern E-bikes have adopted some necessary technology to address these problems, but primarily focus on marketable features instead of safety. Consequently, we strive to meet these demands in a comprehensive manner by developing an affordable and compact bicycle system that accommodates traditional bikes, providing signaling and other support for the rider. Our interdisciplinary experience in embedded firmware development, analog and digital electronics, PCB design, power systems, sensor integration, and mechanical packaging aims to closely reflect real-world development workflows used in consumer electronics industries [SSS5].

2.2 Objectives and Goals

2.2.1 Basic Goals

- Develop a bicycle-mounted safety system to improve rider visibility, safety, and communication with nearby drivers and pedestrians.
- Create an automatic brake light system using IMU-based deceleration sensing.
- Implement left and right turn signals using handlebar controls.
- Improve rider awareness by detecting approaching vehicles and providing warning alerts
- Design a compact, battery-powered system that mounts underneath a bicycle seat.
- Integrate sensing, processing, power management, and L.E.D control into a single working prototype.

2.2.2 Basic Objectives

- Configure the microcontroller for system control and state management.
- Interface the IMU using I2C communication for deceleration detection.
- Activate brake light L.E.Ds when measured deceleration exceeds a threshold.
- Implement left and right turn signals using handlebar switches and MOSFET-driven L.E.Ds.
- Design regulated 3.3V and 5V supplies powered by a rechargeable battery.
- Integrate a radar sensor capable of detecting approaching vehicles and triggering an audio alarm when a vehicle is within a predefined range.

2.2.3 Advanced Goals

- Improve braking detection accuracy by filtering out road vibration and unwanted sensor noise.
- Increase turn signal responsiveness using interrupt-driven inputs.
- Design efficient L.E.D driver circuitry for high-brightness daytime visibility.
- Develop a rechargeable and power-efficient embedded system with custom PCB integration.
- Reduce false brake light activation caused by bumps or sudden movement.
- Increase the accuracy and usefulness of vehicle detection by reducing false alerts and improving hazard awareness.
- Improve system reliability and durability during outdoor riding conditions.

2.2.4 Advanced Objectives

- Implement hardware debounce filtering and interrupt-driven input handling for reliable turn signal activation.
- Apply filtering techniques to reduce IMU noise and prevent false brake activation.
- Design high-current L.E.D driver circuitry with PWM-based brightness and flashing control.

- Design and fabricate a compact custom PCB for sensing, control, and power circuitry
- Integrate lithium-ion charging circuitry into the power subsystem.
- Implement filtering and threshold-based processing of radar data to distinguish legitimate approaching vehicles from stationary objects and environmental noise.

2.2.5 Stretch Goals

- Implement auto-canceling turn signals
- Improve visibility through additional adaptive lighting features, such as automatically adjusted L.E.D brightness
- Improve rider awareness by adding a handlebar-mounted display for vehicle detection information

2.2.6 Stretch Objectives

- Integrate a light sensor and dynamically adjust L.E.D brightness
- Automatically deactivate turn signals after a specified time or after detecting a completed turn using IMU data
- Integrate a small LCD or OLED display to show radar-based vehicle distance, warning status and/or turn signal status

2.3 Existing Technologies and Related Projects

Listed in the chart are some existing technologies that aim to improve rider visibility and safety on bicycles. Using the idea behind these existing products, we aim to combine the features into one system that ensures rider safety.

2.3.1 Table of existing technologies

Product	Features	Cons/Comparison
Garmin RTL515	Radar Detection and Rear Lighting	No turn signal
Unit 1 Smart Light and Remote and helmet	Turn signal, rear and front light and hud overlay	Requires an expensive helmet
CYCL WingLights for bicycle	Turn signal	No rear light or brake light

2.3.2 Garmin RTL515

The first product we have listed is the Garmin RTL515, a bicycle safety device that combines a rear light with radar detection to alert riders of vehicles

approaching from behind. The system improves rider awareness and visibility, but it does not include built-in turn signals(Garmin and subsidiaries *Varia™ RTL515/516*). Our project differs by focusing on both rider visibility and rider communication through integrated brake lights and turn signals.

2.3.3 Unit 1 Smart Light and Remote + Helmet

The second product we have listed is the UNIT 1 Smart Light system, which includes front and rear lighting, turn signals, and a heads-up display integrated into a smart helmet. While the system offers several advanced safety features, it requires the rider to purchase and use a specialized helmet(“Best Bike”), which increases the overall cost. In comparison, our project is designed to be a more affordable add-on system that can be mounted onto a traditional bicycle.

2.3.4 CYCLE WingLights

The third product we have listed is the CYCL WingLights system. This technology uses handlebar-mounted turn signal lights, which helps cyclists communicate turns and lane changes more clearly (“WingLights”). Although the system improves rider signaling, it does not include a rear brake light or automatic braking detection. Our design expands on this idea by combining turn signals, rear lighting, and automatic brake light activation into a single integrated safety system.

2.4 Hardware/Software Design Specifications

The table below represents the system specifications for the B.E.A.C.O.N project. Each specification is outlined with its own range of target values, and method of verification. Most importantly, the highlighted rows within the table represent the three selected specifications intended for verification purposes with the final design. These traits have been agreed upon as the best features to represent the B.E.A.C.O.N project and its capabilities.

2.4.1 Overall system Specifications Table

Specification	Target Value	Verification Method
Break Detection Response Time	150 - 200 ms	Inertia Measurement Unit Timing
Turn Signal activation Latency	50 - 100 ms	Interrupt Timing Test
Break Detection Accuracy	At least 93%	Controlled Breaking Trials

Visible Signaling Distance	At least 300 ft	Outdoor Field Testing
Battery Runtime	At least 8 hours	Battery Discharge Testing
Wireless Communication Range	At least 10 ft	Range Testing
System Operating Voltage	3.3 - 5V	Multimeter Test
Weather Resistance	At least Rain Resistant	Water test
Maximum weight	At most 1 lb	Scale Measurement
Radar Detection Accuracy	At least 90%	Controlled Radar Testing

2.4.2 Component-Level Specifications Table

Component	Specification	Target
IMU Sensor	Accelerometer Range	± 16 g
IMU Sampling Rate	Data Acquisition Frequency	100 Hz
L.E.Ds	Drive Current	350 mA
Microcontroller	Clock Frequency	80 MHz
Battery	Capacity	2500 mAh
Charging IC	Charge Current	500 mA
Wireless Module	BLE Version	5.0
MOSFET Driver Efficiency	Power Efficiency	90%
PCB Dimensions	Maximum Size	100 mm x 60 mm
Radar	Radar Range	at least 100 ft

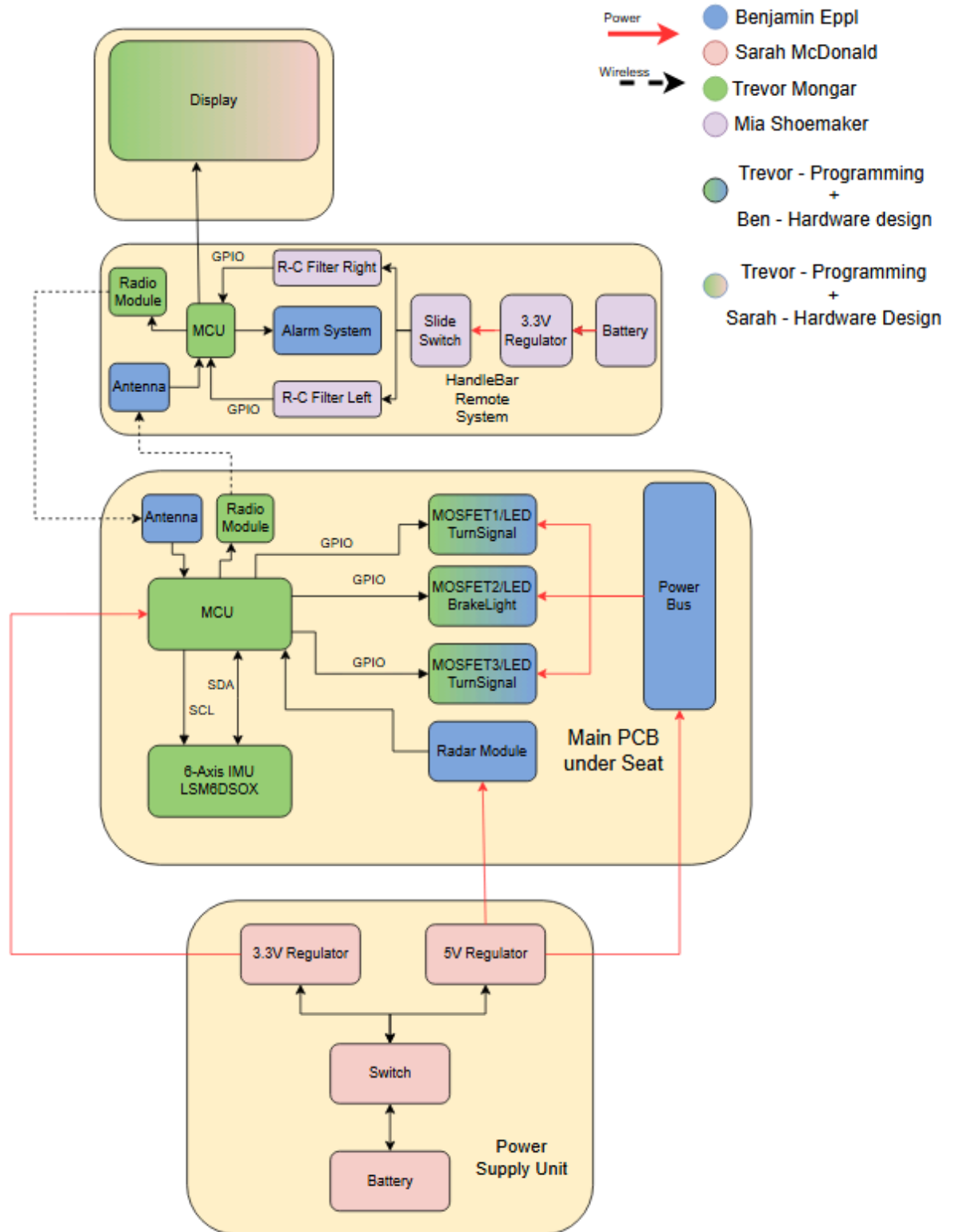


Figure 2.6.1 Hardware Diagram by Authors

2.7 House of Quality

Pictured below is the house of quality diagram used for this project. This diagram provides a visual aid of the engineering characteristic tradeoffs. The far left column shows common customer requirements (WHATs) versus the second row which represents the technical requirements (HOWs). By referencing this diagram, future decisions can be made towards the project on what features could be traded off or which are crucial features.

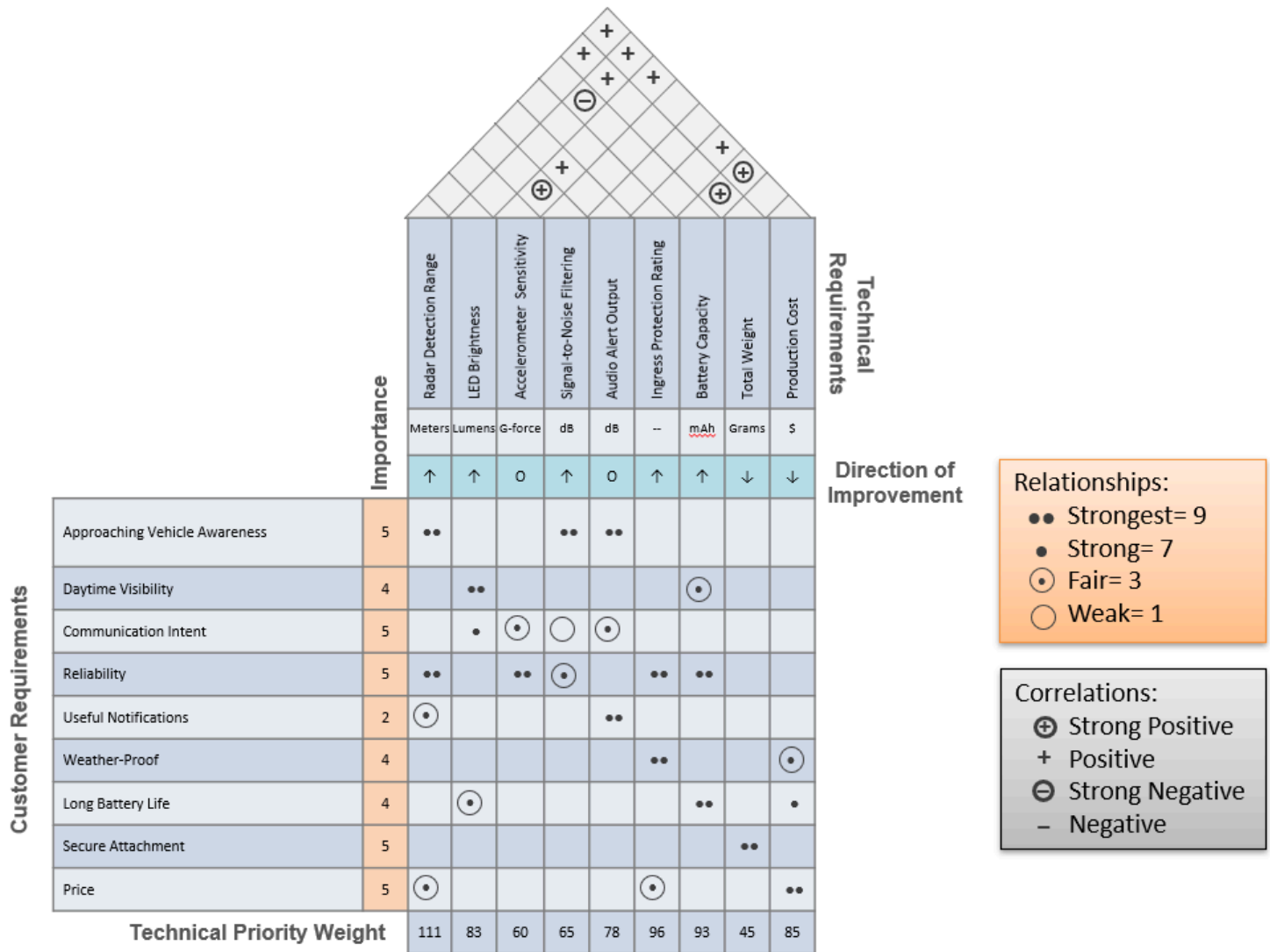


Figure 2.7.1 House of Quality Diagram

2.8 Prototype Renders

Below you can see the prototype based on an imagined final design concept.



Figure 2.8.1 Complete View

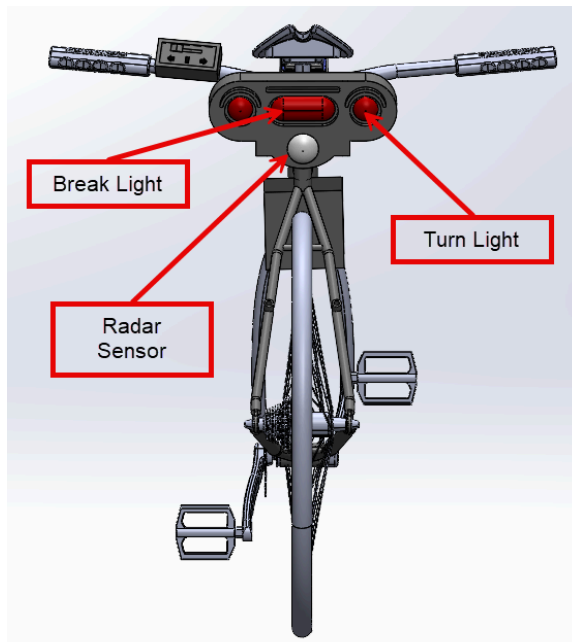


Figure 2.8.2 Back view

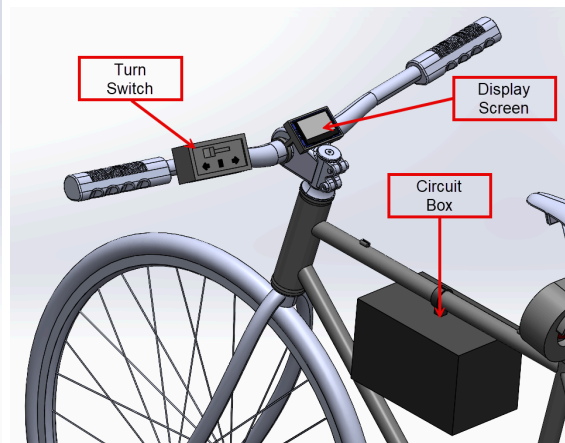


Figure 2.8.3 Front view

3.0 Research and Investigation

3.1 Embedded Processing Hardware

3.1.1 Embedded Processing Hardware Technology Section

The Beacon system requires processing hardware to collect data from sensors, control the different system components, and communicate between the handlebar controller and the rear signaling pod. These devices will be responsible for processing sensor inputs, monitoring rider controls, and controlling outputs such as the turn signals, brake light, warning indicators, and optional display. To determine the best processing solution for the project, three common embedded technologies were researched and evaluated: microcontrollers (MCUs), field-programmable gate arrays (FPGAs), and single-board computers (SBCs). Each technology offers different strengths in areas such as processing power, power consumption, expandability, and overall cost. The following sections provide an overview of each technology before comparing them to determine which is the best fit for this project.

3.1.1.1 MCUs

A microcontroller (MCU) is a small computer built onto a single integrated circuit that is designed to perform specific control tasks in electronic systems. Instead of requiring separate components for processing, memory, and input/output functions, an MCU combines these features into one compact chip. Because of this, microcontrollers are commonly used in embedded systems, where they are programmed to monitor inputs, process information, and control connected hardware.

A typical microcontroller contains a central processing unit (CPU), memory, and input/output (I/O) interfaces. The CPU executes the programmed instructions, while the memory stores both the program and temporary data needed during operation. The input and output interfaces allow the microcontroller to communicate with sensors, switches, displays, motors, and other electronic devices. During operation, the MCU continuously receives input signals, processes the data according to its program, and produces outputs to control the system.

Microcontrollers are used in a variety of embedded applications, including household appliances, automotive systems, industrial equipment, medical devices, and consumer electronics. They are well suited for systems that require reliable, automated control of specific functions. Since many embedded devices operate on batteries, microcontrollers are often designed to consume very little power while maintaining efficient performance.

A microcontroller could be used to collect information from sensors, process that information, and control outputs such as warning lights, alarms, or displays. Some advantages of using an MCU include its compact size, low cost, low power consumption, and ability to integrate multiple functions on a single chip. However, microcontrollers have limited processing power and memory compared to larger computing platforms, making them best suited for applications that perform dedicated control tasks rather than complex or intensive processing [MMM11].

3.1.1.2 FPGAs

A field programmable gate array (FPGA) is a type of integrated circuit that can be programmed after it has been manufactured. Unlike chips that are designed for only one purpose, an FPGA can be reconfigured to perform different hardware functions depending on the needs of the application. This flexibility allows the same device to be reused for different designs or modified as system requirements change.

An FPGA is made up of many programmable logic blocks that are connected through configurable routing paths. Instead of running instructions one after another like a traditional processor, the logic blocks are configured to create a custom digital circuit that performs the required operations directly in hardware. Engineers typically describe the desired hardware using a hardware description language (HDL), and software tools convert that design into a configuration that programs the FPGA.

FPGAs are commonly used in applications that require high performance, fast data processing, or hardware that may need to be updated over time. They are often found in areas such as telecommunications, aerospace, data centers, artificial intelligence, scientific research, and hardware prototyping. Since an FPGA can be reprogrammed without changing the physical chip, it is useful for developing new systems or testing designs before creating custom hardware.

For the Beacon project, an FPGA could be programmed to process data from multiple sensors and control different hardware functions. Some advantages of FPGAs include their flexibility, ability to perform many operations at the same time, and the option to reconfigure the hardware if the design changes. However, programming an FPGA requires specialized design tools and hardware description languages, making development more complex than many other embedded systems [MMM12].

3.1.1.3 SBCs

A single-board computer (SBC) is a complete computer built onto a single circuit board that contains a processor, memory, storage, and input/output interfaces. Unlike a traditional desktop computer, all of the essential components are integrated onto one board, creating a compact system that is capable of running

a full operating system. This allows SBCs to perform more complex computing tasks while still maintaining a relatively small size.

An SBC operates much like a standard computer by running an operating system that manages software and hardware resources. It receives input from connected devices, processes the data using its processor and memory, and produces outputs through its available interfaces. Because SBCs can run operating systems such as Linux, they are capable of multitasking and supporting a wide range of software applications.

Single-board computers are commonly used in education, robotics, home automation, industrial systems, Internet of Things (IoT) devices, and software development. Their compact design, processing capability, and flexibility make them suitable for applications that require more computing power than a typical embedded controller. They are also frequently used for prototyping and developing new hardware and software systems.

For the Beacon project, an SBC could be used to process sensor data, communicate with external devices, and control system outputs. Some advantages of using an SBC include its high processing capability, ability to run a full operating system, compact size, and support for a wide variety of software applications. However, these capabilities also result in higher power consumption and greater software complexity compared to simpler embedded processing platforms, making SBCs better suited for applications that require advanced computing rather than dedicated control tasks [MMM13].

3.1.2 Embedded Processing Unit Technology Selection

3.1.1.1 Embedded Processing Unit Technology Comparison Table

Criteria	MCU	FPGA	SBC
Processing Power	Moderate	Very High	High
Power Consumption	Low	Moderate/High	High
Development Difficulty	Easy	Difficult	Moderate
Hardware Flexibility	Moderate	Very High	Moderate
Cost	Low	High	Moderate
Best Suited for	Embedded control	Parallel	Advanced

	and sensor interfacing	processing and custom digital logic	computing and multitasking
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Based on the comparison in table above, microcontrollers were selected as the processing technology for the Beacon project. Although FPGAs provide excellent performance for parallel processing and SBCs offer greater computing capability, these features exceed the requirements of the project while increasing power consumption, cost, and system complexity. Microcontrollers provide sufficient processing capability, low power consumption, and simple integration with sensors and peripheral devices, making them the most appropriate choice for this project.

3.1.3 Embedded Processing Unit Part Selection

3.1.3.1 ESP32-S3-WROOM-1

The ESP32-S3-WROOM-1 was considered as a potential microcontroller for the Beacon project because it combines wireless communication, processing capability, and a wide variety of peripheral interfaces into a single module. These features make it suitable for embedded systems that require communication with sensors, user inputs, displays, and other electronic components while remaining compact and relatively inexpensive. Since overall project cost is an important design consideration, using a module that integrates many functions may also reduce the need for additional hardware.

The ESP32-S3-WROOM-1 is built around a dual-core 32-bit Xtensa LX7 processor operating at speeds of up to 240 MHz. It includes integrated 2.4 GHz Wi-Fi and Bluetooth 5 Low Energy (LE), along with built-in flash memory and optional PSRAM depending on the module configuration. The microcontroller also supports up to 36 GPIO pins and a variety of communication interfaces, including UART, SPI, I²C, I²S, USB, PWM, ADC, and CAN. These interfaces allow the module to connect with sensors, buttons, L.E.Ds, displays, and other peripherals that may be required in an embedded system. For a bicycle safety device such as the Beacon project, these capabilities could support sensor data collection, rider inputs, warning indicators, and wireless communication between system components.

One advantage of the ESP32-S3-WROOM-1 is that it combines processing, wireless communication, and peripheral interfaces into a single module. This simplifies hardware design and can reduce the number of external components needed, helping lower both system complexity and overall cost. The module also supports low-power operating modes, making it suitable for battery-powered applications where energy efficiency is important. Its large number of GPIO pins

and communication interfaces also provide flexibility for integrating additional hardware if future system requirements change.

The ESP32-S3-WROOM-1 does have several limitations. Because wireless communication and multiple peripherals share the processor's resources, careful software design is required to maintain reliable performance. The module also operates from a 3.3 V supply, so additional circuitry may be needed when interfacing with components that require different voltage levels [MMM14].

3.1.3.2 MSP430FR6989

The MSP430FR6989 was considered as a potential microcontroller for the Beacon project because it is designed for low-power embedded applications and provides a variety of built-in peripherals for interfacing with sensors and other electronic components. The device combines processing, memory, and communication interfaces into a single chip while maintaining very low power consumption. Since battery life and overall project cost are important design considerations, the MSP430FR6989 was evaluated as a possible solution for managing system inputs and outputs.

The MSP430FR6989 features a 16-bit RISC processor that operates at clock speeds of up to 16 MHz. It includes up to 128 KB of nonvolatile FRAM and 2 KB of SRAM, along with multiple communication interfaces such as UART, SPI, and I²C. The microcontroller also provides analog-to-digital converters (ADCs), timers, PWM outputs, GPIO pins, and other peripherals commonly used in embedded systems. These features allow the device to communicate with sensors, buttons, L.E.Ds, displays, and other hardware that may be required in a bicycle safety system such as the Beacon project.

One advantage of the MSP430FR6989 is its ultra-low-power design, which makes it well suited for battery-powered applications that require long operating times. The integrated FRAM provides fast, low-energy memory access while also allowing frequent data writes with high endurance. The device also offers a wide range of built-in peripherals, allowing many embedded applications to be implemented without requiring significant additional hardware. These characteristics can simplify system design while helping reduce overall power consumption and system cost.

The MSP430FR6989 does have several limitations. Compared to higher-performance microcontrollers, it operates at a lower clock speed and provides less processing capability for computationally demanding applications. The device also does not include built-in Wi-Fi or Bluetooth connectivity, requiring additional external hardware if wireless communication is needed [MMM15].

3.1.3.3 STM32F446RE

The STM32F446RE was considered as a potential microcontroller for the Beacon project because it offers high processing performance and a wide range of built-in peripherals for embedded applications. The device combines a powerful processor with multiple communication interfaces, memory, and input/output capabilities on a single chip. Its broad peripheral support and flexible hardware options make it a candidate for systems that require communication with multiple sensors and electronic devices. Since processing capability, expandability, and overall project cost are important design considerations, the STM32F446RE was evaluated as a possible solution for the Beacon system.

The STM32F446RE features a 32-bit Arm Cortex-M4 processor operating at speeds of up to 180 MHz. It includes up to 512 KB of Flash memory and 128 KB of SRAM, along with a wide range of communication interfaces including UART, SPI, I²C, USB, CAN, and I²S. The microcontroller also provides multiple timers, PWM outputs, analog-to-digital converters (ADCs), digital-to-analog converters (DACs), and numerous GPIO pins for connecting external hardware. These peripherals allow the device to interface with sensors, buttons, L.E.Ds, displays, and other components that may be required in a bicycle safety system such as the Beacon project.

One advantage of the STM32F446RE is its high processing performance while still maintaining relatively low power consumption for an embedded microcontroller. Its large memory capacity and extensive peripheral support provide flexibility for integrating multiple hardware components without requiring significant additional circuitry. The device also supports several low-power operating modes, making it suitable for applications where energy efficiency is important. These features allow the STM32F446RE to support both current system requirements and potential future expansion.

The STM32F446RE does have several limitations. Unlike some modern microcontrollers, it does not include built-in Wi-Fi or Bluetooth connectivity, so wireless communication requires additional external hardware. The large number of available peripherals and configuration options can also increase software complexity and development time [MMM16].

3.1.4 Comparison of MCUs

3.1.4.1 Comparison of MCUs Table

Criteria	ESP32-S3-WRO OM-1	MSP430FR6989	STM32F446RE
Processor	Dual-core Xtensa LX7, up to 240 MHz	16-bit RISC, up to 16 MHz	ARM Cortex-M4, up to 180 MHz

Flash Memory	Up to 16 MB	Up to 128 KB	Up to 512 KB
SRAM	Up to 512 KB	Up to 2 KB	Up to 128 KB
Communication Interfaces	UART, SPI, I ² C, I ² S, USB, CAN	UART, SPI, I ² C	UART, SPI, I ² C, USB, CAN, I ² S
Power Consumption (estimate)	80 to 240 mA	100 to 300 microA	20 to 40 mA
Cost (estimate)	\$8 to \$15	\$4 to \$10	\$8 to \$15
Comparable Features	Built-in Wi-Fi/Bluetooth, dual-core, touch sensing	Ultra-low power, FRAM memory	Cortex-M4 with FPU, DAC, advanced timers

Based on the comparison in the table, the ESP32-S3-WROOM-1 was selected as the microcontroller for both the handlebar controller and the rear signaling pod. While the MSP430FR6989 offers lower power consumption and the STM32F446RE provides strong processing performance, neither includes built-in wireless communication. Since the Beacon system consists of two controllers that may communicate wirelessly, the ESP32-S3-WROOM-1 simplifies the design by integrating both Wi-Fi and Bluetooth into a single module.

The ESP32-S3-WROOM-1 also offers the highest processing capability of the three MCUs with its dual-core 240 MHz processor, while providing sufficient Flash memory, RAM, and communication interfaces to support the project's sensors and peripherals.. These features allow it to manage the radar, IMU, lighting, user controls, and optional display while maintaining a common hardware platform for both controllers. Using the same microcontroller in both subsystems also simplifies hardware integration, software development, and future system expansion, making the ESP32-S3-WROOM-1 the most suitable choice for the Beacon project.

3.2 Rear Traffic Sensor Research

3.2.1 Rear Traffic Sensor Technology Research

A key feature of the BEACON project is the ability to detect vehicles approaching from the rear and alert the rider. To accomplish this, a rear-mounted sensing module is integrated into the main PCB assembly beneath the bicycle seat. Sensor data is continuously processed by the system microcontroller to identify nearby objects and assess potential collision risks. When a hazardous condition is detected, the system activates visual and audible alerts to increase rider

awareness. Since sensor performance directly affects the accuracy and reliability of the warning system, we researched multiple types of sensors. We want to ensure that we choose the most reliable, accurate, power and cost effective sensor.

3.2.1.1 Ultrasonic Sensors

Ultrasonic sensors operate by transmitting ultrasonic sound waves toward a target and then receiving the reflected echo. The sensor measures the time required for the sound wave to travel to the object and return. Using this time-of-flight measurement, the distance between the sensor and the object can be calculated. The accuracy of the measurement depends on how effectively the sound wave is reflected back to the sensor. Flat surfaces positioned perpendicular to the sensor produce the strongest reflections, while smaller or irregularly shaped objects may produce weaker responses [MMM4].

One of the main advantages of ultrasonic sensors is that their performance is not affected by the target's color. They can measure distances ranging from a few centimeters up to approximately five meters while maintaining high reliability and precision. Ultrasonic sensors also provide rapid measurement updates and consistent results, making them suitable for real-time monitoring and detection applications [MMM4].

Ultrasonic sensors also have several limitations. Their accuracy can be affected by environmental conditions such as temperature and humidity because these factors influence the speed of sound. The sensors may also become less reliable if they become dirty, wet, or frozen. In addition, they cannot operate in a vacuum because sound waves require a medium through which to travel. Their physical size may also make them less suitable for compact embedded systems [MMM4].

Ultrasonic sensors are widely used in industrial and automation systems. Common applications include robotic arm positioning, object detection, proximity sensing, collision avoidance systems, conveyor belt monitoring, and measuring the dimensions or fill levels of objects. Their ability to provide reliable distance measurements without physical contact makes them useful in many sensing applications [MMM4].

While ultrasonic sensors can be used for collision avoidance and object detection, they are not the most suitable choice for detecting approaching vehicles in a bicycle safety system. Their maximum sensing range is typically limited to only a few meters, which provides relatively little warning time for a cyclist. Additionally, ultrasonic sensors only measure the presence and distance of an object; they do not directly provide information about a vehicle's speed or direction of travel.

3.2.1.2 Lidar Sensors

LiDAR (Light Detection and Ranging) is a sensing technology that uses laser light to measure distances and create detailed three-dimensional representations of an environment. By collecting large amounts of spatial data, LiDAR can generate highly accurate maps and models of objects and surroundings. The technology is used across a variety of industries, including construction, engineering, architecture, surveying, and autonomous systems.

LiDAR sensors operate by emitting rapid pulses of laser light and measuring the time required for those pulses to reflect off objects and return to the sensor. Using these measurements, the system calculates distances and creates a collection of spatial data points that can be assembled into a three-dimensional model of the surrounding environment. This process allows LiDAR to capture both the shape and location of objects with a high level of precision [MMM5].

LiDAR is commonly used for creating accurate digital models of physical environments. According to Matterport, applications include as-built documentation, design validation, progress tracking, and the creation of digital twins. The technology is valued in situations where detailed spatial information is needed to analyze, measure, or monitor real-world environments [MMM5].

One of the primary advantages of LiDAR is its ability to produce highly accurate spatial measurements and detailed three-dimensional maps. Because it directly measures distances using laser pulses, it can capture precise geometric information about objects and environments. This high level of accuracy makes LiDAR useful for applications that require detailed measurements and modeling [MMM5].

A limitation of LiDAR systems is that they generally require more complex hardware and data processing than simpler sensing technologies. The large amount of spatial data collected by the sensor must be processed and interpreted to create useful models or measurements. In many applications, this can increase system complexity and computational requirements [MMM5].

LiDAR can be used to detect approaching vehicles because it provides accurate distance measurements and detailed information about the surrounding environment. Its ability to continuously monitor objects and determine their position makes it a considerable option for collision warning and obstacle detection systems. However, identifying whether a vehicle is approaching and determining its relative speed typically requires additional processing of multiple distance measurements over time. Furthermore, the complexity and computational requirements associated with processing LiDAR data may increase the overall system cost and design effort.

3.2.1.3 Continuous Wave (CW) Doppler Radar

Continuous Wave (CW) radar is a type of radar system that continuously transmits a radio frequency signal rather than sending pulses. The radar analyzes the reflected signal returning from a target to determine whether the object is moving and to estimate its relative velocity. Because the transmitted signal is continuous, CW radar is particularly effective at detecting motion and measuring speed through the Doppler effect. It is commonly used in applications where velocity information is more important than precise location information [MMM6].

A CW radar system continuously emits an electromagnetic signal toward a target. When the signal reflects off a moving object, the frequency of the reflected wave changes slightly due to the Doppler effect. By comparing the transmitted signal with the received signal, the radar can determine the frequency shift and calculate the target's relative velocity. The larger the frequency shift, the greater the speed of the moving object. Since the radar transmits continuously without measuring pulse travel time, it focuses on detecting motion rather than determining range.

CW radar is widely used in applications that require speed measurement and motion detection. Common examples include traffic speed monitoring, vehicle speed measurement, motion sensors, intrusion detection systems, and certain navigation and tracking systems. Its ability to quickly detect moving objects makes it useful in situations where velocity information is the primary concern.

One of the primary advantages of CW radar is its relatively simple design compared to more advanced radar systems. Since it continuously transmits a signal, it can provide real-time velocity measurements with high sensitivity to moving targets. CW radar systems are also capable of detecting small frequency changes, allowing them to accurately measure relative motion. Additionally, continuous transmission can simplify the hardware requirements and improve reliability in motion-detection applications.

A significant limitation of CW radar is that it cannot directly determine the distance to a target. Because the radar continuously transmits rather than measuring the travel time of a pulse, it lacks the information needed to calculate range. As a result, while the radar can determine that an object is moving and estimate its speed, it cannot accurately identify how far away the object is. This limitation can make it difficult to distinguish between multiple moving targets located at different distances.

Continuous Wave radar is capable of detecting approaching vehicles and estimating their relative speed, making it a potential candidate for a bicycle safety system. Its ability to quickly identify moving vehicles could provide useful warning information to a cyclist. However, because the sensor cannot directly measure

the distance to a vehicle, it may be difficult to determine the level of risk associated with an approaching object. For example, the system may detect that a vehicle is moving toward the bicycle but cannot easily determine whether the vehicle is several meters away or much closer. While CW radar offers strong motion-detection capabilities, its inability to provide range information limits its effectiveness in applications that require both distance and speed awareness.

3.2.1.4 Frequency Modulated Continuous Wave (FMCW) Radar

Frequency Modulated Continuous Wave (FMCW) radar is a radar system that continuously transmits an electromagnetic signal while varying its frequency over time. Unlike traditional continuous wave radar, which primarily measures velocity, FMCW radar is capable of determining both the distance and relative motion of a target. This capability allows the system to gather more detailed information about objects within its detection range and makes it useful for applications requiring accurate target tracking and positioning.

An FMCW radar continuously transmits a signal whose frequency changes according to a predetermined pattern, commonly referred to as a chirp. When the signal reflects from a target and returns to the radar, it is compared with the currently transmitted signal. Because the transmitted frequency changes continuously, there is a measurable frequency difference between the transmitted and received signals. This frequency difference, known as the beat frequency, is used to calculate the target's distance. Additionally, motion of the target introduces a Doppler frequency shift, allowing the radar to determine the target's relative velocity.

FMCW radar is used in a wide range of sensing and detection applications where information about both distance and motion is required. Common uses include automotive sensing systems, aircraft navigation, industrial monitoring, surveillance systems, and object tracking applications. The ability to simultaneously determine range and velocity makes FMCW radar useful in environments where multiple moving targets may be present.

One of the primary advantages of FMCW radar is its ability to measure both distance and relative velocity using a continuously transmitted signal. This allows the radar to provide more information about a target than systems that measure only motion. FMCW radar also offers high measurement accuracy and efficient use of transmitted power because the signal is continuously present rather than being transmitted in short pulses. Additionally, the technology is capable of detecting and tracking multiple targets within the sensing area.

FMCW radar systems generally require more complex signal processing than simpler radar technologies. Determining range and velocity involves analyzing frequency differences between transmitted and received signals, which increases computational requirements. The design and implementation of FMCW radar

systems may therefore be more complicated than other sensing technologies. In addition, system performance can depend on factors such as signal processing accuracy and the ability to distinguish between multiple reflected signals.

FMCW radar is well suited for detecting approaching vehicles because it can determine both the distance and relative motion of a target. By continuously monitoring objects within its detection range, the system can track vehicle movement and provide information about changes in position over time. The ability to detect multiple targets and distinguish between them can also be beneficial in environments where several vehicles or objects may be present. However, implementing an FMCW radar system typically requires more advanced processing and design considerations than simpler sensing technologies due to the complexity of analyzing reflected radar signals [MMM7].

3.2.2 Sensors Technology Selection

3.2.2.1 Comparison of Radar Sensors Table

Criteria	Ultrasonic	LiDAR	CW/Dopplar Radar	FMCW Radar
Detection Range	Up to 5 meters	Hundreds of meters	Tens of meters	Hundreds of meters
Distance Measurement	Yes	Yes	No	Yes
Velocity Measurement	No	Requires multiple measurements	Yes	Yes
Implementation Complexity	Low	Moderate-high	Low	Moderate
Cost	\$5 to \$20	\$50 to \$500+	\$5 to \$30	\$50 to \$300+

Based on the comparison in the table, FMCW radar was selected as the sensing technology for the Beacon project. While ultrasonic sensors are inexpensive and simple to integrate, they have a limited detection range and are more affected by environmental conditions. LiDAR sensors provide accurate distance measurements but can be impacted by weather and lighting conditions while offering limited speed detection. Doppler radar is effective at detecting motion and measuring speed but cannot accurately determine the distance to an object. FMCW radar combines the advantages of these technologies by measuring both the distance and relative speed of approaching vehicles while maintaining

reliable performance in various environmental conditions. These capabilities make FMCW radar the most suitable sensing technology for providing timely and accurate warnings in the Beacon bicycle safety system.

3.2.3 Rear Traffic Sensor Part Research

3.2.3.1 Infineon BGT60TR13C

The Infineon BGT60TR13C was considered as a potential FMCW radar sensor for the Beacon project because it is designed for short-range object detection and motion sensing while maintaining a compact form factor. The sensor operates in the 60 GHz frequency band and uses frequency-modulated continuous wave (FMCW) radar technology to detect the presence, distance, speed, and direction of moving objects. Its small size and integrated antenna design make it suitable for embedded applications where space is limited.

The BGT60TR13C integrates one transmit antenna and three receive antennas within a single package and supports a bandwidth greater than 5 GHz for high-resolution measurements. The sensor is controlled through an SPI interface and includes an internal finite-state machine (FSM) that manages radar sweeps, data acquisition, and storage. These features allow the sensor to capture radar data while simplifying communication with an external microcontroller such as the ESP32-S3.

One advantage of the BGT60TR13C is its ability to measure multiple target characteristics, including range, speed, angle of arrival, and direction of motion. The integrated antenna-in-package design reduces the amount of external RF circuitry required and simplifies hardware integration. The sensor also supports configurable FMCW operation and optimized power modes, providing flexibility for different embedded sensing applications. These capabilities make the sensor suitable for applications requiring reliable object detection and tracking.

The BGT60TR13C does have several limitations. The sensor communicates raw radar data through its SPI interface, meaning an external processor is responsible for performing signal processing and interpreting the measurements. In addition, configuring FMCW radar parameters and processing radar data requires more software development than simpler distance sensors [MMM8].

3.2.3.2 Acconeer XM125

The Acconeer XM125 was considered as a potential FMCW radar sensor for the Beacon project because it is a compact radar module designed for embedded sensing applications. The module combines Acconeer's A121 60 GHz pulsed coherent radar sensor with an onboard Arm Cortex-M4 microcontroller, allowing it to perform radar measurements while communicating with an external host

controller. Its compact size and integrated design make it suitable for applications requiring object detection, motion sensing, and distance measurement.

The XM125 is built around the A121 60 GHz radar sensor and an 80 MHz Arm Cortex-M4 microcontroller with 128 KB of Flash memory and 64 KB of RAM. It supports communication through UART and I²C interfaces, allowing an external microcontroller such as the ESP32-S3 to receive radar information or configure the module. The onboard microcontroller runs Acconeer's Radar System Software (RSS), which manages radar operation and provides a software interface for application development.

One advantage of the XM125 is that it combines the radar sensor, processing hardware, and communication interfaces into a single module, reducing the amount of external circuitry required. The module also supports low-power operation, has a compact 18.6 mm × 15 mm footprint, and can operate as either a stand-alone system or alongside an external host controller. These features simplify hardware integration and make the module well suited for embedded radar applications.

The XM125 does have several limitations. Because it is a complete radar module with an onboard microcontroller, it is generally more expensive than a standalone radar sensor. It is also designed around Acconeer's Radar System Software, which requires the use of provided software framework when developing applications [MMM9].

3.2.3.3 RFbeam V-LD1-RFB-00M-01

The RFbeam V-LD1 was considered as a potential FMCW radar sensor for the Beacon project because it is a compact radar module designed for embedded distance sensing applications. The module operates at 61 GHz using FMCW radar technology and includes integrated signal processing, allowing it to measure the distance of detected objects without requiring complex radar processing on an external microcontroller. Its small surface-mount package also makes it suitable for integration onto a custom PCB.

The V-LD1 combines a radar front end, signal processing hardware, and a digital UART interface into a single module. It supports distance measurements of up to 50 meters and can communicate directly with a host microcontroller such as the ESP32-S3. The module also provides access to additional radar data if more advanced processing is needed.

One advantage of the V-LD1 is that it performs the radar signal processing internally, reducing the amount of software development required on the host microcontroller. Its compact size, low-power operation, and simple UART interface also make it easier to integrate into an embedded system while reducing overall hardware complexity. The V-LD1 does have several limitations.

The module is primarily designed for distance measurement and may require additional software if more advanced radar features or target tracking are added [MMM10].

3.2.4 Selection of FMCW Radar Part

3.1.9 Comparison of FMCW Radars Table

Criteria	Infineon BGT60TR13C	Acconeer XM125	RFbeam V-LD1
Detection Capability	Range, speed, angle, motion	Distance and object detection	Distance measurement
Communication Interface	SPI	UART, I2C	UART
Signal Processing	External processing required	Onboard MCU and radar processing	Integrated signal processing
Integration Complexity	High	Low	Low
Power Consumption	Low	Low	Low
Cost	\$20	\$22	\$50

Based on the comparison in Table 3.1.9, the Acconeer XM125 was selected as the FMCW radar module for the Beacon project. While the Infineon BGT60TR13C offers advanced radar capabilities, it requires significant external signal processing, increasing both software complexity and development time. The RFbeam V-LD1 provides a simple, integrated solution for distance measurement, but it is primarily designed for ranging applications and offers fewer advanced detection capabilities. The XM125 provides a good balance between performance and ease of integration by combining onboard radar processing with simple UART and I²C communication. This allows the ESP32-S3 to receive processed radar information instead of raw radar data, reducing software complexity while still providing the functionality needed for detecting approaching vehicles. As a result, the XM125 was determined to be the most suitable radar module for the Beacon project.

3.3 Internal Sensor Research

3.4 Lithium-Ion Battery Research

A battery is required for B.E.A.C.O.N to function. The project includes a microcontroller, radar sensor, wireless transmitter, display, indicator lights, brake lights, turn signals, and a buzzer for alerts and alarms. Because the device is intended to be mounted on a bicycle, the battery has to be small and lightweight while still being able to store the energy required to power the system for about 12 hours per battery cycle. The battery we choose must be safe for the cyclist, and be somewhat weather proof to operate within normal outdoor temperatures, and remain cost-effective.

3.4.1 Lithium-Ion Battery Technology Research

Lithium-ion batteries are the most common rechargeable battery types used today. They can be seen almost everywhere being in most portable electronic devices, including laptops and iPads. These batteries provide high energy density, low weight, and a constant high number of charge and discharge cycles [BBB11] making them useful for B.E.A.C.O.N, where the battery needs to be compact but still powerful enough to run multiple electronic pcb boards and lights at the same time. The most common lithium-ion types are Lithium Cobalt Oxide (LiCoO_2), Lithium Manganese Oxide (LMO), Lithium Iron Phosphate (LiFePO_4), and Lithium Nickel Manganese Cobalt Oxide (NMC). These different batteries are compared based on their energy density, safety, cost, service life, and discharge efficiency [BBB11] and each one has different projects they would be better for.

Lithium Cobalt Oxide batteries are commonly used in smartphones, tablets, and laptops. This lithium type is popular because it provides an extremely high amount of energy for its size and cost [BBB11]. This battery has a Lithium Cobalt Oxide cathode and a graphite anode. However, this battery has very alarming safety concerns because it has lower thermal stability than other lithium-ion types [BBB11]. For a bicycle-mounted system like B.E.A.C.O.N, thermal stability and safety are very important due to the battery being exposed to outdoor temperatures ranging from hot to cold, vibrations, and possible impacts like crashes or potholes.

Another concern is battery aging. With regular battery use, the repeated charging and discharging causes the Solid Electrolyte Interphase (SEI) layer on the graphite anode to eat away at the lithium in the battery [BBB12]. As it consumes the lithium inside the battery, the battery capacity reduces over time [BBB12]. This means the battery may not hold as much charge and be more unreliable. Since B.E.A.C.O.N is intended for long-term repeated use, battery capacity retention is a deal breaker.

Lithium Manganese Oxide batteries use a manganese-based cathode, usually with a spinel structure [BBB11]. This type of battery has low internal resistance,

allowing it to support high discharge currents [BBB11]. It is also more thermally stable than Lithium Cobalt Oxide batteries, making it safer in outdoor applications [BBB11]. However, Lithium Manganese Oxide batteries do have a lower energy density and shorter cycle life compared to other lithium-ion battery types [BBB11]. For B.E.A.C.O.N, this makes LMO less optimal because this project needs a battery that is small, lightweight, and able to power the system for at least 12 hours.

Next is Lithium Nickel Manganese Cobalt Oxide, or NMC. In recent years, NMC batteries have become much more popular and one of the most widely used lithium-ion battery types because they provide a strong balance between energy density, safety, performance, lifetime, and cost [BBB11]. This battery combines the high energy density of nickel, the thermal stability of manganese, and the performance benefits of cobalt [BBB11]. It's the happy medium between these battery types. For B.E.A.C.O.N, this balance makes NMC a strong option because the battery must be lightweight, reliable, and able to power the system for regular use.

Overall, lithium-ion batteries have the strongest general appeal for B.E.A.C.O.N because they provide higher energy density than Nickel-Metal Hydride batteries while remaining practical for small portable devices [BBB11]. Within the lithium-ion category, NMC was selected because it provides a better balance of energy density, safety, lifetime, and cost than the other lithium-ion chemistries considered [BBB11].

3.4.1.1 Lithium Polymer

Lithium Polymer batteries, also called LiPo batteries, are a variation of standard lithium-ion batteries. The main difference between lithium-ion batteries and Lithium Polymer batteries is the type of electrolyte and packaging they use. Standard lithium-ion batteries usually use a liquid electrolyte inside a cylindrical or rectangular cell. Lithium Polymer batteries use a polymer-based electrolyte and are often made in a flexible pouch-cell form [BBB11]. This pouch design allows the battery to be thin, lightweight, and available in many different shapes and sizes [BBB11].

Lithium Polymer batteries have several advantages. They have high energy density and are light for the amount of energy they can store [BBB11]. Because they can be made in thin pouch shapes, they are commonly used in portable applications where size and weight are important. These applications include drones, RC vehicles, smartphones, and other small electronic devices. For B.E.A.C.O.N, this is useful because the system must remain small enough to mount under a bicycle seat or on the handlebars without adding unnecessary bulk.

LiPo batteries can also provide higher discharge current than some other rechargeable batteries. This makes them useful for applications that need short

bursts of high current. They also have low self-discharge and good efficiency. These characteristics make Lithium Polymer batteries a strong option for many portable systems, especially when weight and size are major design concerns [BBB11].

One example of this type of battery is the LP556877 Lithium Polymer rechargeable battery. This battery has a nominal voltage of 3.7 V, a capacity of 3300 mAh, and dimensions of approximately 77 mm × 68 mm × 5.5 mm [BBB14]. Its thin pouch shape and relatively high capacity show why Lithium Polymer batteries are commonly considered for compact electronic systems [BBB14]. These characteristics would make a LiPo battery attractive for B.E.A.C.O.N if size and weight were the only design concerns.

However, Lithium Polymer batteries also have important disadvantages. Since they are often made in pouch-cell form, they can swell as they age. They are also more vulnerable to punctures and mechanical damage compared to cylindrical battery cells. Because of this, they must be charged, stored, and used carefully. Protection electronics are commonly required to prevent overcharging, over-discharging, and other unsafe battery conditions [BBB11].

This is an important concern for the B.E.A.C.O.N system because the battery will be used in a bicycle-mounted environment. The system may experience vibration, outdoor weather, and occasional impact. A pouch-style battery could be more difficult to protect mechanically in this environment. Although Lithium Polymer batteries are lightweight and compact, they may require more careful packaging and protection than cylindrical lithium-ion cells.

Even though Lithium Polymer batteries have high energy density and low weight, they are not the best choice for the B.E.A.C.O.N system. The system will be mounted underneath a bicycle seat, where mechanical durability and safe packaging are important. For this type of environment, cylindrical lithium-ion batteries are more suitable because they are easier to mount securely and protect from damage. For this reason, Lithium Polymer batteries were considered, but an NMC lithium-ion battery solution was chosen instead [BBB11].

3.4.1.2 Lithium Iron Phosphate

Lithium Iron Phosphate batteries, also called LiFePO_4 batteries or IFR batteries, are another type of rechargeable lithium battery. They use a Lithium Iron Phosphate cathode and a graphite carbon anode. Unlike Lithium Cobalt Oxide or NMC batteries, the cathode of a LiFePO_4 battery has an olivine crystal structure [BBB11]. This structure gives the battery strong thermal and chemical stability. Because of this, Lithium Iron Phosphate batteries are considered one of the safer rechargeable lithium battery types [BBB11].

One of the biggest advantages of Lithium Iron Phosphate batteries is their long cycle life. These batteries can go through many charge and discharge cycles

while only losing a small amount of capacity over time [BBB11]. This makes LiFePO_4 batteries useful in applications where reliability and long service life are very important.

Lithium Iron Phosphate batteries also have strong thermal stability. They have a lower risk of thermal runaway, overheating, and combustion compared to many other rechargeable lithium-based batteries [BBB11]. Because of this, they are often used in electric vehicles, backup power systems, and energy storage systems. These are applications where safety and long-term reliability are very important. For B.E.A.C.O.N, this safety advantage is valuable because the device will be used close to the rider.

Another advantage of LiFePO_4 batteries is their high-current discharge capability. This means they can provide a steady voltage even when a large amount of current is needed for a short time [BBB11]. This can be useful in applications that need temporary high-current output and long battery life.

However, Lithium Iron Phosphate batteries also have drawbacks. Their energy density is lower than the energy density of NMC batteries [BBB11]. This means that a LiFePO_4 battery pack would need to be larger and heavier to store the same amount of energy. For small portable systems, this can become a major issue because every additional gram of weight and every extra amount of space matters. Since B.E.A.C.O.N is intended to be mounted on a bicycle, size and weight are major design factors.

Another issue is cell voltage. A typical LiFePO_4 cell has a nominal voltage of about 3.2 V [BBB11]. A two-cell LiFePO_4 pack would provide approximately 6.4 V nominal, which is lower than the 7.2 V to 7.4 V range desired for the B.E.A.C.O.N power system. This may require additional power regulation or a different pack configuration. In comparison, two NMC lithium-ion cells in series can provide approximately 7.2 V nominal, which better matches the system power requirement.

Although Lithium Iron Phosphate batteries are safe, reliable, and long-lasting, they are not the best option for the B.E.A.C.O.N system. The system is mounted underneath the seat of a bicycle, so the battery needs to be as compact and lightweight as possible. A larger and heavier battery pack would make the system less practical for everyday use. For this reason, LiFePO_4 batteries were considered but were not selected. Instead, an NMC battery was chosen because it provides higher energy density and a more suitable two-cell voltage for the system, while still keeping the size and weight manageable [BBB11].

3.4.1.3 Nickel-Metal Hydride

Nickel-Metal Hydride batteries, also called NiMH batteries, are commonly used in many rechargeable applications. They are found in portable electronics, cordless

tools, hybrid vehicles, emergency equipment, and many consumer products [BBB9]. NiMH batteries are popular because they are low cost, reliable, and generally safe to use [BBB8], [BBB9]. In many of these applications, energy density is not the most important factor [BBB9].

One of the main advantages of NiMH batteries is their safety. NiMH batteries do not have the same thermal runaway concerns as some lithium-based chemistries [BBB8]. They also support many charge and discharge cycles and are one of the more mature rechargeable battery technologies [BBB8]. Since this battery type has been available for a long time, NiMH batteries are easy to find from local and online retailers. This makes them a cost-effective option for many types of rechargeable battery applications.

NiMH batteries are also considered more environmentally friendly than some older rechargeable batteries. For example, they do not contain cadmium, which is found in older Nickel-Cadmium batteries [BBB9]. Cadmium can be hazardous during transport, storage, or disposal, so the use of NiMH batteries helps reduce this concern [BBB9]. For these reasons, NiMH batteries remain popular in applications where safety, cost, and reliability are important.

However, NiMH batteries also have several disadvantages, especially for small portable systems. The biggest issue is their low energy density [BBB9]. This means that a NiMH battery pack would take up more space and weigh more than a lithium-ion or Lithium Polymer battery pack with a similar capacity. NiMH batteries also have higher self-discharge, meaning they lose stored energy faster when they are not being used [BBB8]. This makes them less ideal for systems that may sit unused for long periods of time.

Another disadvantage is that a single NiMH cell only has a nominal voltage of about 1.2 V [BBB8]. Since the B.E.A.C.O.N system requires approximately 7.2 V to 7.4 V, a NiMH battery pack would need about six cells connected in series. This would make the battery pack larger, heavier, and more difficult to package. In comparison, an NMC lithium-ion battery pack can reach a similar voltage using only two cells in series.

Even though NiMH batteries offer good safety, reliability, and low cost, they are not the best choice for the B.E.A.C.O.N system. A NiMH battery pack would be larger and heavier than other battery options with similar energy storage. Since B.E.A.C.O.N must remain compact and lightweight for bicycle use, NiMH batteries were ruled out. For this reason, Lithium Nickel Manganese Cobalt Oxide batteries were chosen instead [BBB8], [BBB9].

3.4.2 Lithium-Ion Battery Technology Selection

3.4.2.1 Lithium-Ion Battery Technology Comparison Table

Battery Type	Energy Density	Weight	Safety	Cycle Life	Cost
NMC	High	Low	Medium	500 - 1000	Low
Lithium Polymer	Very High	Very Low	Medium Low	300 - 800	Medium
Lithium Iron Phosphate	Medium	Medium	Very High	2000 - 5000	High
Nickel-Metal Hydride	Low	High	Medium High	500 - 1000	Low

For the B.E.A.C.O.N project, an NMC lithium-ion battery was selected because it provides a strong balance of energy density, weight, safety, and cost [BBB11]. The system needs a battery that can power the microcontroller, radar sensor, wireless communication module, display, lights, and buzzer while still being compact enough to mount on a bicycle. NMC chemistry meets these needs better than the other battery types that were considered.

3.4.3 Lithium-Ion Battery Part Research

3.4.3.1 Part 1 Panasonic NCR18650GA

This battery was very close to being selected for BEACON's battery. NCR1860GA meets all of the requirements for a battery; 3.6 V nominal voltage, 3350 mAh capacity, 48 g weight. A 2 pack of this battery would be a great choice for the 7.2 V battery pack needed for this project but the datasheet was less detailed than the Samsung one. This battery will be used as the back up plan if our group encounters any unforeseen issues with the Samsung battery. [BBB16]

3.4.3.2 Part 2 INR18650 MJ1 3500mAh

This battery also met all of the specifications with a 3.635 nominal voltage, 3500 mAh capacity, and weighs 49 g. This battery has the best specifications with the highest capacity and highest discharge rating but is the most expensive option. [BBB17]

3.4.3.3 Samsung INR18650-35E

The Samsung INR18650-35E is another strong candidate with a nominal voltage of 3.6 V, 3350 mAh capacity, and a weight of 50 g. This battery is an extremely common battery that has been used in many applications and has a very readable cut sheet. [BBB13]

3.4.3.4 Lithium-Ion Battery Technology Comparison Table

Battery	Nominal Voltage	Weight	Capacity	Max Discharge Current	Cost
Samsung INR18650-35E	3.6 V	50 g	3350 mAh	8 A	\$14
Panasonic NCR18650GA	3.6 V	48 g	3350 mAh	10 A	\$17
LG Chem INR18650 MJ1	3.635 V	49 g	3500 mAh	10 A	\$26

3.4.4 Lithium-Ion Battery Part Selection

The selected battery cell is the Samsung INR18650-35E. This cell uses NMC chemistry and provides a nominal voltage of 3.6 V, a nominal capacity of 3350 mAh, and a maximum continuous discharge current of 8 A [BBB13]. When two of these cells are connected in series, the battery pack provides 7.2 V and about 24 Wh of stored energy [BBB13]. This gives the system enough power while keeping the battery pack small and lightweight.

The Samsung INR18650-35E is also a cylindrical 18650 cell, which makes it easier to mount and protect compared to a pouch-style Lithium Polymer battery. A cylindrical cell will be placed into an enclosed battery pack, which will be easily used on a bicycle-mounted system that may experience vibration and movement. This makes the selected cell more practical for the physical design of B.E.A.C.O.N.

The Panasonic and Samsung batteries were the toughest 2 to choose from with how similar the specifications from the cut sheets are; but the Samsung was chosen due to its slightly lower cost and how it is the most common battery that can be easily troubleshooted online if any problems arise.

Overall, the NMC lithium-ion battery was chosen because it provides the best balance for the B.E.A.C.O.N system. Lithium Polymer batteries are lightweight but are more vulnerable to swelling and mechanical damage [BBB11]. Lithium Iron Phosphate batteries are very safe and long-lasting but are heavier and have lower energy density [BBB11]. Nickel-Metal Hydride batteries are safe and low cost but would require more cells and create a larger battery pack [BBB8], [BBB9]. The NMC lithium-ion battery provides enough energy in a compact form, making it the most practical choice for a bicycle-mounted safety system [BBB11], [BBB13].

3.5 Antenna Research

3.5.1 Antenna Technology Research

A wireless communication link is required for the B.E.A.C.O.N system because the project contains three separate electronic subsystems but two will be connected wirelessly. The first subsystem is mounted underneath the bicycle seat and contains the main sensing, lighting, radar, and power circuitry. The second subsystem is mounted on the handlebar and is used for rider input and display interaction. These two subsystems must be able to communicate the turn signal commands, system status information, and rider warning information; an antenna is required for wireless communication between the two boards. The current B.E.A.C.O.N specifications identify Bluetooth Low Energy 5.0 as the wireless communication method and require a minimum communication range of 10 ft.

Since Bluetooth Low Energy operates in the 2.4 GHz ISM band, the antenna used for this project must support 2.4 GHz wireless communication. According to the Bluetooth Low Energy physical layer specification, Bluetooth LE operates from 2400 MHz to 2483.5 MHz and uses 40 RF channels spaced 2 MHz apart [BBB1]. Therefore, the selected antenna must operate efficiently across this frequency range while remaining compact enough to fit within the bicycle-mounted enclosures.

For the B.E.A.C.O.N project, the antenna must also be low cost, durable, easy to integrate into the PCB design, and reliable in the expected mounting environment. Since the system will be placed near a bicycle frame, battery pack, wiring, and enclosure materials, antenna placement is an important design consideration. Conductive materials, nearby ground planes, and poor antenna clearance can reduce wireless performance. Because of this, the selected antenna solution should require as little custom RF design as possible while still meeting the short 10 ft Bluetooth Low Energy communication requirement.

Several antenna options were considered for the B.E.A.C.O.N system. These included the integrated PCB antenna on the ESP32-S3-WROOM-1 module, an external flexible PCB antenna, a ceramic chip antenna, and a custom PCB trace antenna. Each option has advantages and disadvantages related to size, cost, layout complexity, and expected wireless performance. For this project, the integrated antenna on the ESP32-S3-WROOM-1 module was selected as the best option because it is already included with the selected microcontroller module and is sufficient for the required short-range Bluetooth Low Energy link [BBB2].

3.5.1.1 Integrated PCB Antenna

The integrated PCB antenna is the preferred antenna option for the B.E.A.C.O.N system. The selected ESP32-S3-WROOM-1 module already includes a PCB antenna, which reduces the amount of RF design required by the team [BBB2]. This is important because antenna design can be sensitive to PCB layout, enclosure materials, nearby conductors, and the physical placement of the module. By using a module that already includes an antenna, the design avoids the need to create, tune, and test a custom antenna from the beginning.

The ESP32-S3-WROOM-1 module supports 2.4 GHz wireless communication, which matches the Bluetooth Low Energy frequency range required for B.E.A.C.O.N [BBB1], [BBB2]. Since the project only requires a minimum communication distance of 10 ft between the under-seat subsystem and the handlebar subsystem, the integrated antenna is expected to provide enough range for normal operation. This makes it a practical choice for a short-range bicycle safety system.

Another advantage of the integrated PCB antenna is that it reduces the number of additional components needed on the PCB. No separate antenna, coaxial cable, U.FL connector, or external antenna mounting location is required. This helps reduce cost, assembly time, and mechanical complexity. It also reduces the chance of connector or cable failure caused by bicycle vibration.

However, the integrated antenna still requires proper module placement. The antenna end of the ESP32-S3-WROOM-1 should be placed near the edge of the PCB, and the antenna area should not be covered by copper, large metal components, the battery pack, or the bicycle frame. Espressif's PCB layout guidance shows that antenna clearance and module placement are important for maintaining proper RF performance [BBB3]. If the antenna is placed too close to metal or if the clearance area is not respected, the wireless range and reliability could be reduced.

For B.E.A.C.O.N, the integrated PCB antenna provides the best balance of simplicity, cost, and expected performance. It supports the required 2.4 GHz Bluetooth Low Energy communication, does not require extra RF hardware, and keeps the PCB design simpler than the other antenna options. Because of these advantages, the ESP32-S3-WROOM-1 integrated antenna was selected as the primary antenna solution for the project [BBB2], [BBB3].

3.5.1.2 External Flexible PCB Antenna

Another antenna option considered for B.E.A.C.O.N was an external flexible PCB antenna. A flexible PCB antenna is a thin antenna printed on a flexible substrate. These antennas are commonly used for 2.4 GHz wireless applications such as Wi-Fi, Bluetooth, Zigbee, and other ISM-band devices [BBB6]. They are usually connected to a wireless module using a small coaxial cable and an antenna connector, such as a U.FL or I-PEX connector [BBB6].

The main advantage of an external flexible PCB antenna is placement flexibility. Since the antenna is connected with a cable, it can be positioned away from the main PCB, battery pack, metal bicycle frame, or other objects that may reduce RF performance. This could be useful if the under-seat enclosure or nearby bicycle hardware interferes with the integrated antenna. A flexible PCB antenna could be mounted on the inside of a plastic enclosure or placed in a better location to improve wireless performance [BBB6].

Flexible PCB antennas are also thin and lightweight, which makes them useful for compact electronic products. Many flexible antennas include adhesive backing, allowing them to be mounted inside an enclosure without taking up much space [BBB6]. For B.E.A.C.O.N, this could be useful if the project later needs more control over antenna placement or if testing shows that the integrated antenna does not provide reliable communication between the handlebar and under-seat subsystems.

However, external flexible PCB antennas also have disadvantages. They require an external antenna connector, a coaxial cable, and a physical mounting location for the antenna. These additions increase cost, assembly time, and mechanical complexity. They also introduce more possible failure points, especially in a bicycle-mounted system that will experience vibration and movement.

Using an external flexible PCB antenna would also require changing the selected module. The ESP32-S3-WROOM-1 includes an integrated PCB antenna, while the ESP32-S3-WROOM-1U version is designed for use with an external antenna connector [BBB2]. Since the current design uses the ESP32-S3-WROOM-1, an external antenna is not needed unless range testing shows that the integrated antenna does not meet the project requirement. For this reason, the external flexible PCB antenna is considered a backup option rather than the primary antenna solution [BBB2], [BBB6].

3.5.1.3 Ceramic Chip Antenna

A ceramic chip antenna was also considered for the B.E.A.C.O.N system. A ceramic chip antenna is a small surface-mount antenna component that can be placed directly on a PCB. These antennas are commonly used in compact wireless devices where board space is limited, including Bluetooth, Wi-Fi, Zigbee, and IoT products [BBB5]. Their small size makes them useful when the designer needs a smaller antenna solution than a printed PCB trace antenna.

The main advantage of a ceramic chip antenna is its compact physical size. This could be useful for the handlebar-mounted subsystem, where the available PCB area may be limited. A ceramic chip antenna can fit into small wireless designs and may allow more flexibility in the mechanical layout of the board [BBB5].

However, ceramic chip antennas are not simple drop-in components. They still require careful PCB layout, proper antenna placement, a ground clearance or keep-out region, and a matching network to operate correctly [BBB5]. The antenna performance can be affected by the PCB size, ground plane, enclosure material, nearby components, and the final product environment. If the layout or matching network is not designed correctly, the antenna may become detuned, which can reduce wireless range and reliability [BBB5].

For the B.E.A.C.O.N project, a ceramic chip antenna is not the preferred solution because the selected ESP32-S3-WROOM-1 module already includes an integrated PCB antenna [BBB2]. Using a separate ceramic chip antenna would require changing to a bare ESP32-S3 design or using a module designed for an external antenna. This would increase PCB complexity and require additional RF design work without providing a major benefit for the required 10 ft Bluetooth Low Energy link. Because the integrated antenna already meets the needs of the project, the ceramic chip antenna was considered but not selected [BBB2], [BBB5].

3.5.1.4 Custom PCB Trace Antenna

Custom PCB trace antennas were also evaluated. A custom PCB trace antenna is an antenna integrated within a PCB that you build yourself. This option could be low cost if it does not require a separate antenna component. Customs add a much broader range of choices with inverted-F antennas, meandered antennas, and monopole-style trace antennas. A custom PCB trace antenna can be useful in high-volume products where reducing component cost is important; but this project does not need this advantage.

There are multiple downsides with this antenna type. One disadvantage is that it requires careful RF design and adds to the scope of the project. The scope would now have to factor in, antenna geometry, board material, ground plane, matching circuit, nearby components, and enclosure can all lead to performance and accuracy questions. With the 2.4 GHz frequency that will be used for this project, small layout changes can cause impedance mismatch, signal loss, and reduced range. Also the ESP32-S3 hardware design guidance states that the antenna area should be placed carefully and kept away from copper, routing, and nearby components to preserve RF performance [BBB3].

For the B.E.A.C.O.N device, a custom PCB trace antenna is not the best option because the project does not require a long-range or custom-shaped antenna. The selected ESP32-S3-WROOM-1 already includes a PCB antenna, so designing a new custom antenna would add unnecessary risk [BBB2]. Since this project is being developed as a senior design prototype, reducing RF design risk is more important than optimizing antenna cost for large-scale production.

3.5.2 Antenna Technology Selection

3.5.2.1 Antenna Technology Comparison Table

Antenna Type	Advantages	Disadvantages	Best fit for Project
Integrated PCB antenna on ESP32-S3-WROOM-1	Minimal design complexity, no extra antenna part, compact, already included with selected module	Requires correct module placement and antenna clearance. Less leeway in placement.	Best choice for project
External flexible PCB antenna	Can be placed away from metal, battery, and enclosure interference	Requires WROOM-1U module, U.FL/I-PEX connector, cable, and extra cost	Backup option
Ceramic chip antenna	Very compact surface-mount option	Requires matching network and careful RF layout	Not optimal
Custom PCB trace antenna	Low component cost and can be printed directly on PCB	Requires RF design, tuning, and testing	Not optimal

3.5.3 Antenna Part Research

3.5.3.1 Taoglas FXP72.07.0053A

The Taoglas FXP72.07.0053A is a strong external antenna option because it is designed for 2.4 GHz Bluetooth, Wi-Fi, Zigbee, and ISM applications. Its datasheet lists 2400–2500 MHz operation, 67.9% efficiency, -1.68 dB average gain, 3.06 dBi peak gain, 50 Ω impedance, and a U.FL-compatible connector. But, since we only need a short range it wouldn't make sense to use an external antenna connector like this one [BBB18].

3.5.3.2 Part 2 Johanson Technology 2440AT62B0085002U

The Johanson 2440AT62B0085002U is the ceramic chip antenna designed for 2.4–2.485 GHz operation. Its specifications are a 50 Ω impedance, 2.1 dBi typical peak gain, 55% typical radiated efficiency at 2.44 GHz, and a 12 × 12 × 4 mm SMD package. But this part is far more complicated due to it using a dual-feed RHCP design and Johanson recommends a 3 dB hybrid coupler when used with a single-feed chipset which adds another component to worry about. This will not

be used since it is far more complex than the other antennas to implement. [BBB19].

3.5.3.3 integrated ESP32-S3-WROOM-1 antenna

The ESP32-S3-WROOM-1 integrated PCB antenna remains the best option for B.E.A.C.O.N. because the selected microcontroller already includes the antenna on the module, this makes the most sense since using another antenna would require a new microcontroller because there is not a port to add a new antenna since this one is integrated. This antenna will save a lot of space and will work excellent for the short range we require the PCBs to communicate over. [MMM14]

3.5.4 Antenna Part Selection

3.5.4.1 Antenna Part Comparison Table

Antenna	Frequency	Gain / Efficiency	Size	Complexity	Cost
ESP32-S3-WROOM-1 built-in antenna	2.4GHz	1 dBi 90%	18.0 x 25.5 x 3.1 mm	Low	free
Taoglas FXP72.07.005 3A	2400 MHz	3.06 dBi 67.9%	31 x 30.4 x 0.2 mm	High	\$4.29
Johanson 2440AT62B00 85002U	2400 MHz	2.1 dBi 55%	12 x 12 x 4 mm	Medium	\$0.53

For the B.E.A.C.O.N project, the integrated PCB antenna on the ESP32-S3-WROOM-1 module is the best option because it supports the required 2.4 GHz Bluetooth Low Energy communication while keeping the design simple, compact, and cost-effective [BBB1], [BBB2]. Since the project only requires communication between the under-seat subsystem and the handlebar subsystem that will be way under the maximum distance of 10 ft, a separate external antenna is not needed at all.

The integrated antenna also reduces the amount of RF design work required for the project. External flexible PCB antennas and ceramic chip antennas can provide useful benefits in some designs, but both options require additional layout considerations, components, and testing. A flexible PCB antenna would require the ESP32-S3-WROOM-1U module, an antenna connector, and a coaxial cable [BBB2], [BBB6]. A ceramic chip antenna would require careful RF layout,

antenna clearance, and a matching network [BBB5]. These added requirements would increase design complexity without a clear benefit for the current B.E.A.C.O.N communication range.

Because of this, the ESP32-S3-WROOM-1 integrated PCB antenna is the most practical choice for the project. It is already included with the selected module, supports the required Bluetooth Low Energy frequency range, reduces PCB complexity, and avoids the need for external RF components. During system testing, our group will verify that the antenna provides reliable communication between the two subsystems. If range or reliability issues occur, the external flexible PCB antenna option will be the backup solution [BBB2], [BBB3], [BBB6].

3.6 Indicator Light Research

One of the necessary components of the B.E.A.C.O.N system is the set of indicator lights used to communicate a cyclist's intentions to other road users. These indicators serve as a visual means of communication, helping nearby motorists, cyclists, and pedestrians to recognize actions such as braking or turning [SSS8]. By providing clear and timely signals, the system aims to improve rider visibility and reduce uncertainty on the road, ultimately contributing to safer interactions between cyclists and surrounding traffic.

This section seeks to identify the possible lighting technologies available for these indicators and determine which of them is best suited for the project. The primary goal of these components is to mimic the signal functionality found in traditional motor vehicles, while adapting them for practical use on a bicycle. Before evaluating each of the potential technologies, it is important to establish the key characteristics required to meet the project's design objectives.

The first requirement is high visibility; these lights must be clearly visible at long distances, during both daytime and nighttime operations. To remain consistent with bicycle lighting requirements adopted by numerous U.S. states, the system should be seen from a minimum distance of 500 ft (152 m), providing adequate warning to approaching motorists [SSS9]. In addition to distance, the indicators should maintain visibility across a wide viewing angle. The signals should be observable not only directly behind the cyclist, but also from adjacent lanes and other off-axis positions. Visibility will also require responsiveness from the light, as longer delays can lead to increased risk for those on the road.

Beyond visibility, the selected technology must also be capable of operating reliably in real-world cycling conditions. Since the B.E.A.C.O.N system will be mounted on a bicycle, the indicator lights will be exposed to constant vibration, weather conditions, and frequent use. As a result, durability and lifespan are important factors when evaluating potential options. The chosen solution should be resistant to mechanical shock and maintain consistent performance over extended periods without requiring frequent maintenance or replacement. This

includes the need for the technology to be economical, as higher prices will cumulate in total costs and impact not just development but consumer demand. Additionally, compatibility with the system's microcontroller and power supply must be considered, as simpler control circuitry can reduce overall system complexity, cost, and power consumption.

With these requirements established, it is necessary to examine the lighting technologies available for use in the system. Several different technologies exist that could be used, such as light-emitting diodes (L.E.Ds), incandescent bulbs, and fluorescent lighting. Each technology offers unique advantages and disadvantages in terms of visibility, efficiency, reliability, and cost. The following sections compare these options and evaluate their suitability for integration into the B.E.A.C.O.N system.

3.6.2 Indicator Light Technology Research

3.6.2.1 *Light Emitting Diodes*

Light Emitting Diodes (L.E.Ds) are semiconductor devices that produce light through a process known as electroluminescence. When a forward bias voltage is applied across a p-n junction, electrons and holes recombine within the semiconductor material, releasing energy [SSS10]. Unlike other bulbs, L.E.Ds do not rely on heating filaments to generate light, resulting in significantly higher electrical efficiency and reduced heat production [SSS11]. In modern automotive, bicycle, and traffic signaling applications, L.E.D technology has become the dominant lighting solution.

The primary requirement for the B.E.A.C.O.N system is visibility. High-brightness red L.E.Ds are capable of producing luminous intensities ranging from several hundred mcd, for standard indicator L.E.Ds, to several candela for high-power signaling. Typical indicator L.E.Ds operate at currents between 10 mA and 30 mA, while higher-power signaling L.E.Ds may operate at several hundred milliamps [SSS12]. Even at these higher currents, L.E.Ds consume substantially less power than incandescent or halogen alternatives while producing comparable or greater light output. This higher brightness, while consuming relatively little power, makes them particularly well suited for battery-powered applications.

L.E.Ds also have extremely fast response times, typically measured in microseconds [SSS13]. In practical applications, the delay introduced by an L.E.D is effectively negligible when compared to the processing and sensing delays of the microcontroller and inertial measurement unit (IMU). This allows brake indications to activate almost immediately following the detection of a deceleration event, providing motorists with the earliest possible warning.

Additionally, L.E.Ds are highly resistant to vibration and mechanical shock [SSS14]. Because L.E.Ds are solid-state devices with no moving parts or fragile filaments, they exhibit excellent resistance to vibration and mechanical shock. This characteristic is particularly important for bicycle-mounted applications. Their operational lifespan also allows for over 25,000 to 50,000 hours of use, making them one of the longest lasting options available.

The primary disadvantages of L.E.Ds include the need for current limiting circuitry and potential thermal management requirements for high-power devices. Excessive current can damage the device or reduce its lifespan, making proper driver design essential [SSS13]. Additionally, standard bulbs are traditionally priced higher than other alternatives, sitting at around \$3 to \$8 [SSS15]. However, these concerns are generally minor when compared to their benefits. Due to their efficiency, durability, long lifetime, and ease of integration with microcontrollers, L.E.Ds represent a strong candidate for implementation within the system.

3.6.2.2 Incandescent Lightbulbs

The next technology to review is Incandescent Lightbulbs. These bulbs produce light by passing electrical current through a tungsten filament. As the current flows through this filament, moving electrons collide with the tungsten atoms, transferring their energy caused by the resistance of the material. This transfer causes the tungsten to heat to very high temperatures, usually exceeding 2,000° C, which results in lights glowing through a process called incandescence [SSS16]. Since the filament is delicate, it requires protection from the outside environment to facilitate the incandescent radiation that it operates on. For much of the twentieth century, incandescent lighting served as the primary technology for automotive, household, and other lighting applications due to its simple construction and ease of operation.

In terms of visibility, incandescent bulbs are capable of producing sufficient brightness for signaling applications. Historically, they were widely used in automotive brake lights and turn signals before the adoption of L.E.D technology. However, incandescent bulbs achieve this brightness at the cost of significantly higher power consumption. A traditional 25 W bulb incandescent bulb can output over 220+ lumens, which is sufficient for braking systems on the road [SSS17]. A large percentage of the electrical energy supplied to the filament is converted into heat and infrared light, rather than visible light, resulting in poor overall efficiency sitting around 1.3-2%. For battery-powered systems such as B.E.A.C.O.N, this increased power consumption would reduce operating time and place greater demands on the battery subsystem.

Response time is another important consideration for signaling applications. Because incandescent bulbs rely on heating a filament to produce light, they require a short amount of time to reach full brightness after power is applied.

Although this delay is typically measured in tens of milliseconds (about 50ms until discernible optical output), it remains significantly slower than the near-instantaneous response of L.E.Ds. In brake indication systems, even small delays can reduce the amount of warning time available to following motorists and have an important impact on cognitive reaction time [SSS18].

Durability presents an additional concern. The tungsten filament within an incandescent bulb is inherently fragile and susceptible to damage from vibration, shock, and repeated thermal cycling. Since bicycle-mounted devices experience continuous vibration from road surfaces, the likelihood of filament degradation or failure is increased compared to stationary applications. Incandescent bulbs also possess shorter operational lifetimes than most modern lighting technologies, often requiring replacement after only about 750 to 2,000 hours on average [SSS19].

One advantage of incandescent bulbs is their simplicity. Unlike other lighting options, incandescent bulbs do not require specialized current-regulation circuitry and can often be connected directly to an appropriately rated voltage source.

One of the most notable advantages of an incandescent bulb is its pricing; its simplistic design only costs around \$1 to \$3 depending on packaging. While incandescent bulbs are able to provide adequate visibility, their relatively high-power consumption, slower response time, limited lifespan, and reduced durability make them less suitable for the B.E.A.C.O.N system.

3.3.2.3 Fluorescent Lamps

A fluorescent lamp is a mercury vapor lamp that uses the phenomenon of fluorescence to produce visible light. The lamp contains a cathode that allows a voltage to be applied, creating an electrical field, and enabling a current to flow through the tube. A low-pressure mercury vapor contained within a glass tube, which gets ionized. The discharge of mercury atoms produces ultraviolet radiation which in turn excites the phosphor powder coating to radiate visible light [SSS20]. This design allows for a filament-less structure which requires extremely high temperatures to operate, allowing for greater efficiency.

In terms of visibility, fluorescent lighting is capable of producing substantial luminous output. A typical compact fluorescent lamp (C.F.L) operating at approximately 5–7 W can produce between 250 and 450 lumens, while larger fluorescent tubes can exceed several thousand lumens depending on their size and operating conditions [SSS21]. With a traditional small-scale bulb operating at about 900 lumens and 13 watts, outpacing technologies like traditional incandescent bulbs. This benchmark meets the necessary safe operating requirements for daytime visibility.

Compact fluorescent bulbs convert a greater portion of electrical energy into visible light than incandescent bulbs, resulting in efficiencies typically ranging from 8–15%. The lifespan of fluorescent lamps typically ranges from 7,000 to 15,000 hours, depending on factors such as usage conditions and how often they are turned on and off.

Durability is an important concern with C.F.L. Frequent cycles can shorten lifespan as the electrodes located within the lamp degrade over time due to repeated starting cycles. Considering the need for indicator lights in the B.E.A.C.O.N system to turn on and off frequently, this poses a risk to the longevity of the bulb [SSS22]. Furthermore, the presence of small amounts of mercury within the lamp introduces environmental concerns since fluorescent lamps are constructed from thin glass tubes that are vulnerable to damage from impact and vibration, an issue not found in L.E.D alternatives. This design results in a cost ranging from \$2 to \$5.

Response time is another important consideration for signaling applications. Fluorescent lamps generally require a brief startup period before reaching full brightness. Depending on the lamp type, ballast design, and ambient temperature, illumination may occur almost immediately or may require several hundred milliseconds to several seconds to achieve full output. Compact fluorescent lamps are particularly susceptible to slower startup times and reduced performance in colder environments. This delay makes fluorescent technology less suitable for applications requiring immediate visual indication, such as bicycle brake signaling systems.

While fluorescent lamps offer improved efficiency and longer service life than incandescent bulbs, their slower startup characteristics, increased system complexity, sensitivity to vibration, and mercury content make them less suitable for the B.E.A.C.O.N system. Modern L.E.D technology provides superior efficiency, durability, response time, and reliability for bicycle-mounted signaling applications.

3.6.2 Lighting Technology Selection

After reviewing each of the prominent available technologies for the B.E.A.C.O.N system, before a final decision can be made, it is necessary to compare each option against each other.

3.6.2.1 Lighting Technology Comparison Table

Specification	L.E.D	C.F.L	Incandescent Lightbulb
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Response Time (seconds)	~0.000001	≥0.1	~0.050
Lifespan (hours)	25,000 - 50,000	7,000 - 15,000	750 - 2,000
Durability	High	Low	Low
Power Efficiency (lumens per watt)	90 - 150	50 - 70	10 - 17
Cost (usd)	3 - 8	2 - 5	1 - 3

Based on the details provided above for incandescent, compact fluorescent (C.F.L), and light-emitting diode (L.E.D) technologies, L.E.Ds are determined to be the most suitable lighting solution for the B.E.A.C.O.N system. While all technologies evaluated are capable of producing sufficient illumination for signaling applications, L.E.Ds demonstrated the strongest overall performance when compared against the project's design requirements.

Although L.E.Ds typically have a higher initial cost than incandescent and halogen alternatives and may require current regulation of circuitry, these disadvantages are outweighed by their superior efficiency, durability, and operational lifespan. As a result, L.E.D technology offers the best balance of performance, reliability, and power efficiency for the intended application.

3.6.3 Indicator Light Part Research

With L.E.Ds chosen as the technology to be used, it is necessary to then determine which of the many different parts available on the market are best suited to our needs. As mentioned previously, to align with the design objectives of B.E.A.C.O.N, the L.E.D must be visible from a wide angle and at longer distances while prioritizing lower costs and better durability. To mimic similar signaling lights in motor vehicles, it should be colored and have the capacity to adjust its brightness, not just turn on or off.

3.6.3.1 Lumileds LXM2-PD01-0040

The Lumileds LXM2-PD01-0049 is a red high-brightness L.E.D from Lumileds' LUXEON Rebel Color Line. According to the manufacturer's datasheet, this part has a dominant wavelength in the 620 to 645 nm range and is tested at 350 mA, where it produces approximately 40 to 48 lumens depending on the flux bin. That places it in a practical range for a compact signaling source because it provides a clearly defined red output without requiring extreme drive current. The device is intended for a high-power SMD implementation, which makes it suitable for a

compact PCB-based lighting module rather than a bulky lamp-style assembly [SSS23].

For the B.E.A.C.O.N brake light, this L.E.D is a strong candidate because red remains the most intuitive and widely recognized color for braking, and the spectral range of this part allows it to be an option. The moderate flux output is useful for battery-powered operation since it allows for a more flexible balance between brightness and power demand instead of sacrificing efficiency for high minimum operating specifications and a higher maximum output. In a bicycle application, however, a single emitter may not be enough on its own for strong daytime visibility, so this part would likely perform best when paired with a wide optic, reflective housing, or multiple L.E.Ds in a clustered arrangement. Together, these features make this component a good baseline brake light option: efficient, compact, and color-appropriate, while still leaving room for optical enhancement if the final enclosure needs more visual intensity in bright sunlight [SSS23].

3.6.3.2 Cree XLamp XP-E2 Photo Red

The Cree XLamp XP-E2 family is a compact point source L.E.D line that is available in several colors, including red, photo red, amber, and far red. Cree describes the XP-E2 as an upgrade over the original XP-E, with up to 20% higher lumen output, low thermal resistance as, as much as 1.4 °C/W, a wide viewing angle of 105° to 140°, and a maximum drive current of 1.5 A for the color versions. In the XP-E2 datasheet, the photo red bins are grouped around a peak wavelength of roughly 650–670 nm at 350 mA, which places them in the deep-red range commonly used for high-visibility signal lighting. For a brake or rear signal light, the main advantage of the XP-E2 family is its optical control. Because it is a single die point source with a relatively narrow physical footprint, it is easier to shape the beam with a lens or reflector than with a broader diffuse emitter. That is important for a bicycle system, where the light must be noticeable from the rear without wasting too much power, and the wide viewing angle also helps the light remain visible when the bike is seen from off-center positions, which is useful in traffic. The tradeoff is that Cree positions this family as a general lighting and specialty-lighting platform rather than a dedicated signaling platform, so it is very capable electrically and optically, but it is less purpose-built for signaling than an L.E.D family designed specifically around turn-indicator use. Even so, the XP-E2 Photo Red remains a strong choice when compact size, thermal performance, and efficient red output are priorities [SSS24].

3.6.3.3 ams OSRAM OSOLON SSL 120, GA CSSPM1.23

For the turn-light function, the OSRAM amber option is the best fit of the three because amber is the standard signaling color for turn indicators and is specifically associated with visibility and driver recognition. OSRAM's own turn indicator guidance describes turn signals as flashing amber or yellow lights used to support safety and visibility in traffic. The OSOLON SSL 120, GA CSSPM1.23 is

an amber L.E.D in the 609 to 624 nm range, and OSRAM lists it at about 100 lm, 30 cd, 0.74 W, with a 120° beam angle and an operating temperature range of -40 °C to 125 °C. That combination makes it a very practical candidate for an outdoor bicycle signal that needs to remain visible in daylight while still staying efficient [24].

The broader OSOLON family is also designed with signaling-like performance in mind: OSRAM describes OSOLON Signal L.E.Ds as compact, high-performance signaling parts with a 3 mm × 3 mm footprint, a 120° lens design for improved light extraction, and high electro-optical efficiency at elevated temperatures. Those family traits matter for a bike turn lamp because the light must be noticeable from multiple viewing angles, survive vibration and weather, and maintain output without excessive heat buildup. In practical terms, this makes the OSRAM amber part the strongest match for daytime turn indication among the candidates considered here. Its higher luminous output gives it a better chance of cutting through ambient daylight than a lower output indicator L.E.D, and the broad beam helps ensure that the signal is visible not only directly behind the bike but also from angled approaches. For a turn signal, that balance of color, output, beam spread, and efficiency is exactly what the design needs [SSS25].

3.6.4 Lighting Part Selection

After researching several high-brightness L.E.Ds suitable for bicycle signaling applications, selection narrowed to three candidates: the Lumileds LXM2-PD01-0040, Cree XP-E2 Photo Red, and OSRAM OSOLON SSL 120 Amber. These L.E.Ds were chosen due to their high luminous output, relatively low power requirements, and suitability for outdoor signaling applications. Since the B.E.A.C.O.N system is intended to provide brake and turn signal functionality on a bicycle, the most important factors in our evaluation were daytime visibility, power efficiency, viewing angle, operating temperature range, and cost.

3.6.4.1 Light Part Comparison Table

Specification	Lumileds LXM2-PD01-0040	Cree XP-E2 Photo Red	OSRAM OSOLON SSL 120 Amber
Luminous Flux (lm)	~46	~45-55	~100
Forward Voltage (V)	2.9	2.1	2.25
Efficacy (lm/W)	~45	~60	~71
Viewing Angle (°)	125	130	120

Max Current (A)	0.7	1.5	1.0
Operating Temp (°C)	-40 to 125	-40 to 150	-40 to 125
Single Unit Price (\$)	5.33	5.14	1.62

Among the three options, the Lumileds LXM2-PD01-0040 offered a luminous flux of approximately 46 lumens with an efficacy of 45 lm/W. While its 125° viewing angle provides good visibility from multiple directions, it exhibited the lowest efficiency of the candidates considered. Additionally, its forward voltage of 2.9 V and unit cost of \$5.33 make it the most expensive option while simultaneously providing the lowest light output [SSS23].

The Cree XP-E2 Photo Red demonstrated improvements over the Lumileds option in several areas. With a luminous flux between 45 and 55 lumens and an efficacy of approximately 60 lm/W, it provides comparable brightness while consuming less power. The L.E.D also features the widest viewing angle at 130° and the broadest operating temperature range of -40°C to 150°C, indicating excellent reliability in harsh environmental conditions. However, the Cree device has the highest maximum current rating of 1.5 A, potentially increasing thermal management requirements [SSS24].

The OSRAM OSLON SSL 120 Amber emerged as the strongest overall performer. Producing approximately 100 lumens, it delivers nearly twice the luminous output of the other L.E.Ds while maintaining the highest efficacy at approximately 71 lm/W. This combination of brightness and efficiency is especially important for ensuring visibility during daylight operation while preserving battery life. Although its viewing angle of 120° is slightly narrower than the alternatives, it remains sufficiently wide for bicycle signaling applications. Additionally, its operating temperature range of -40°C to 125°C satisfies the environmental requirements of the project. The most significant advantage of the OSRAM L.E.D is its cost, with a unit price of only \$1.62, making it substantially more affordable than the competing options. Therefore, the OSRAM OSLON SSL 120 Amber offers the best overall balance of performance, efficiency, and affordability among the L.E.Ds evaluated [SSS25].

3.7 Communication Protocol Research

Data transmission between hardware components and peripherals is achieved with communication protocols. Each protocol sets the standard, or rules, for how data communication is achieved between devices. For hardware and electronic design, there are several communication protocols, each of which can affect data transmission speed, hardware complexity, or interfacing. For real-world scenarios, there are many properties to consider, such as data transmission

speed and accuracy. For the BEACON project, which contains multiple hardware components and peripherals, each communication protocol must be considered to ensure proper hardware cooperation and operation.

3.7.1 Communication Protocol Technology Research

3.7.1.1 UART

UART (or Universal Asynchronous Receiver Transmitter Protocol) is a universal serial communication protocol that can be applied to any transmitter/receiver. Designed for communication between microcontrollers/microprocessors and the external environment. Additionally, UART is asynchronous, meaning that a clock signal is not shared between devices, and that each side of communication has its own clock source. This can simplify hardware complexity (specifically the wiring) but requires that each communicating participant agree to a communication speed, or a baud rate. It is typical for UART communication to be used for one-way communication between two devices, where the communicating devices are either the “controller” or “peripheral”.

UART communication is highly beneficial for its one-way communication between the microcontrollers and their respective peripherals. UART is typically the industry standard for straightforward data transfers that prioritize data flow architectures. Because of its asynchronous nature, UART communication has a low pin/wire count, requiring only two wires for its transmitting and receiving ends [TTT1]. While there are more efficient options, such as I2C and SPI (discussed later), UART communication is an excellent choice due to its simplicity, low cost, and easy implementation.

However, there are significant drawbacks to consider when using the UART communication protocol. Because of its asynchronous design, UART communication is more susceptible to clock inaccuracies between communicating devices [TTT2]. Clock frequency inaccuracies can lead to data inconsistencies (e.g., missing bits, corrupted data, etc.), significantly affecting the communication between two devices. Additionally, UART has a lower maximum transmission speed in comparison to other communication protocols [TTT2].

For the BEACON project, the UART protocol could be used for communication between the underseat microcontroller and the FMCW radar module. Because the radar module includes its own digital signal processing unit for target velocity calculations, the asynchronous nature of UART communication will allow the radar module to stream data packets back to the microcontroller without concern of clock synchronization. The main concern with this implementation is the slower communication speed between the two. Since UART has a slower maximum transmission speed, hazards detected by the radar sensor could be delayed. Additionally, data from the radar sensor could be corrupted or lost entirely due to any clock desynchronizations.

3.7.1.1 I2C

Inter-Integrated Circuit, also known as Two-Wire Interface, is another serial communication protocol. I2C is a low-speed communication protocol intended for communication between controllers (microcontrollers/microprocessors) and target devices [TTT3]. However, unlike the UART communication protocol, I2C allows for communication between multiple devices. Additionally, I2C is synchronous. Devices communicating over the I2C protocol share a clock reference and use it for transmitting/receiving data.

I2C is designed for short distance, chip-to-chip data communication on a single circuit board, and can be extended across multiple boards in a closed system [TTT4]. Because of the simple setup and hardware integration, I2C is a cheap and simple communication protocol to implement. Additionally, I2C communication offers several different speed modes for data transmission. The fastest data transmission mode, Ultra-Fast mode, can transfer data at speeds of up to 5 megabits per second [TTT3]. However, the faster the data transmission speed, the more inaccurate the data becomes. During communication, an extra bit is included inside the data stream. This extra bit is the ACK/NACK (Acknowledge/Not Acknowledge) bit and indicates whether the transmission was received [TTT3]. The ACK/NACK bit improves the error handling capabilities of the I2C communication protocol.

Like the UART communication protocol, the I2C protocol is on the lower end of the communication protocol speed spectrum. While I2C contains different speed modes to compete with faster protocols, the increased speed reduces data accuracy. In addition to the slower speeds, I2C is restricted to short-distance, on-board communication. As wire distance increases, so does the bus capacitance, limiting data transfer speed and increasing the effect of signal noise. In general, when implementing the I2C communication protocol in an embedded project, overall hardware design needs to be carefully considered. Changes to the I2C implementation (e.g., wire distance, clk speed) can significantly affect performance.

When considering the I2C protocol for the BEACON project, it is clear that components that are low-bandwidth, on-board, and in close proximity to an MCU will benefit the most from this communication protocol. The primary component that comes to mind is the 6-axis IMU, which records acceleration for the brake lights. Due to the speed/accuracy trade-off with I2C, the bandwidth requirement for data transmission must be considered. An IMU accelerometer register read at a frequency of 52-104 Hz [TTT5], which, in fast-mode for I2C, at 400 kHz, can be done in 162.5 μ s, giving the MCU plenty of time to activate the brake light from the accelerometer data. However, not only does bandwidth need to be considered, but so does wire distance. If the IMU component is placed too far

from the controller MCU, data corruption can occur. Therefore, the IMU on-board position needs to be considered when designing the schematic and PCB.

3.7.1.1 SPI

Serial Peripheral Interface, is a synchronous serial communication protocol, like I2C. It enables high-speed data transfer between controller and one or more peripheral devices. Unlike I2C however, the SPI protocol is not standard, and terminologies might differ depending on manufacturer [TTT6]. This is caused by the overall simplicity of the SPI communication protocol and the absence of upper-level protocols [TTT6]. Each implementation can have custom-defined particularities, along with application implemented upper-level protocols [TTT6]. Additionally, SPI enables full-duplex communication. Unlike I2C, which only allows for half-duplex. Meaning, that data can be sent and received at the same time during SPI communication [TTT6].

Arguably the greatest advantage of SPI communication is the data transmission speed. SPI offers high data transmission rates, averaging at 10 MHz [TTT7]. Furthermore, since SPI is a full-duplex communication protocol, data can be sent both ways without having to wait for the other, increasing performance speed. Like I2C, SPI has a simple implementation, requiring only four primary lines (MISO, MOSI, SCLK, and SS). Additionally, despite having four primary data lines, SPI also has a low pin count. The SS line (Originally called “Slave Select” the naming convention for the peripheral devices has changed due to a shift to more inclusive language. Despite that, the name for the SS line stays the same) is used to select the controller device to communicate with. In SPI, each controller device can use its own controller line, reducing the total pin count on the controller device. When the controller device wants to communicate with a specific controller device, the specific SS line needs to be pulled low to enable the selected device. When the controller is done communicating, the line is pulled high, and ends the communication [TTT7].

The SPI protocol also has its downsides. Similar to the I2C protocol, SPI has a limited distance, and is typically used for communication on the same board, or closely located components. Of the same token, the further the peripheral device is from the controller, the more susceptible it is to signal degradation and noise due to the high-frequency signals and lack of differential signaling [TTT7]. Furthermore, due to the continuous, high frequency, clock signals, SPI can lead to higher power consumption compared to other protocols [TTT7]. Devices running the SPI communication protocol need to prioritize efficient power consumption to accommodate.

Components that require high-bandwidth data throughput are more suitable for SPI communication. Aside from that, SPI should be used sparingly, to avoid significant power consumption. For the BEACON project, SPI would be very beneficial for the heads-up display along with the wireless handlebar remote. For

the intended screen for the project, a frequency of 400 kHz would provide a 1 second refresh rate. Making a glorified powerpoint presentation, instead of a dynamic display. If the faster modes of I2C were to be used instead, then the communication would be susceptible to data corruption or loss. Using the high clock speed of SPI will allow the handlebar MCU to update the display at speeds of 30 frames per second. The other primary usage for SPI communication is the wireless turning signal switch. Located on the handlebar system, the turning signal switch needs to maintain an over-the-air latency chain of less than a millisecond, for smooth operation. Since the SPI offers both high-speeds and full-duplex data transmission, it will serve as the perfect communication protocol between switch and MCU. Furthermore, since both the display and the turning signal switch are located in close proximity to the controller, signal degradation and noise will not be an issue. However, because SPI will be integrated into the handlebar remote system, it must prioritize energy efficiency, operating in low-power modes to conserve energy. Otherwise, the remote will run out of power while in use faster than desired.

3.7.2 Communication Protocol Technology Selection

3.7.2.1 Communication Protocol Technology Comparison Table

Protocol	UART	I2C	SPI
Communication Type	Asynchronous	Synchronous	Synchronous
Data Transfer Mode	Full-Duplex	Half-Duplex	Full-Duplex
Number of Wires	2 (TX/RX)	2 (SCL/SDA)	4 (MOSI, MISO, SCK, SS)
Number of Devices	2 (between 2 devices)	Multi-Controller and Multi-Peripheral	One controller with many peripheral
Speed	Low	Medium-High	High
Use Case	Serial communication between two devices	Short-range, multi-device communication	Fast communication for real-time processing

For the BEACON project, each of the three listed communication protocols will be implemented. The BEACON project employs a variety of peripherals that each have their own communication requirements. Modern radar sensors employ an on-board digital signal processing (DSP) unit, which is also why radar sensors contain its own independent internal clock. This especially complements UART

communication, due to its asynchronous design. Additionally, the radar's finished calculations are transmitted as data packets that are at most 100-300 bytes per frame. UART communication is capable of handling this data transference with a baud rate of roughly 115,200 bps.

I2C communication will be employed with the 6-Axis IMU. When configured to fast mode (400 kHz), I2C communication will be capable of handling brake detection from deceleration signals sent by the IMU. I2C fast mode is capable of reading the IMU accelerometer data when streamed at high-performance mode of 417 Hz [TTT5]. Furthermore, I2C can be implemented for communication between the MCU and the other peripherals, aside from the radar sensor that is employing UART. Due to I2C's shared bus architecture, a controller can communicate with multiple peripheral devices with each having their own unique, hard-coded, 7-bit address [TTT3]. Of the same token, extra peripherals communicating with the MCU will share the physical communication pins (SDA and SCL), significantly reducing the required wire spacing. I2C is intended as the main form of communication between controllers and peripherals between both systems aside from unique exceptions.

SPI communication will be implemented between the display unit and the handlebar system MCU. SPI communication operates at a frequency range of 10 - 50 MHz, allowing for the display to reach at least a 30 Hz refresh rate. Moreover, SPI offers full-duplex communication. The display and controller can send and receive data simultaneously, allowing for efficient data reading and writing while still making sure the display stays updated. Like I2C, SPI has a simple setup. Requiring only 4 communication pins. Additionally, if further peripheral devices are added to the communication, SPI can be used for multiple peripheral devices at the cost of an additional Slave Select line.

3.8 Software Research

For designing an engineering project, the software involved in production is fundamental to ideal product performance. The software implemented ranges from the programming language used to the integrated development environment. Furthermore, software packages meant for modeling (such as solidworks) or schematic design (such as autodesk) can streamline the design and production process. Alleviating many of the potential issues caused by hand-drawn diagrams. To ensure the least problematic design implementation, many different software and programming languages were considered for the BEACON project.

3.8.1 Programming Languages Research

3.8.1.1 C/Embedded C

The programming language “Embedded C” is a specialized subset and extension of the C programming language. The EmbeddedC (abbreviated to “EC”) is primarily used in the development of Embedded Systems. As of 2025, EC has remained the backbone of Embedded System design [TTT8]. EC has many advantages that reinforce its industry standard status such as its portability, its direct low-level hardware control, and efficiency/optimization for resource-constrained devices [TTT8]. More than any other, the EC programming language has become a staple for embedded development.

Due to EC’s capability of providing low-level hardware access with minimal computational overhead, the language has remained an industry standard. Arguably one of the greatest strengths of EC (and C in general) is that it is a very “low-level” high-level language [TTT9]. EC gives an extensive amount of direct hardware control to the developer, without losing the “higher-level” features. Through mapping software constructs directly to memory-mapped I/O registers, physical features such as hardware interrupt lines, can be directly manipulated with very few penalties to performance [TTT10]. Through the use of dynamic memory addressing, such as pointers, a developer can directly read/write to the physical memory addresses that hardware components use to map their internal control and status registers [TTT9].

The C programming language is a compiled procedural language, and by extension, so is EC. When run through a compiler, the entire file is converted to machine code all at once. Removing the need for any on-chip interpreter and significantly increasing execution times. Furthermore, the EC language does not use a managed runtime environment or garbage collector like other high-level languages. This assures a deterministic performance upon execution [TTT11], allowing for a fixed and predictable number of clock cycles for execution control architecture for every instance.

Because EC offers direct control of hardware, it also introduces potentially severe vulnerabilities in regards to memory security and software stability, among others. Unlike other high-level languages, C/EC does not contain native type safety, memory bounds-checking, or managed exception handling. Furthermore, programming oversights such as unmanaged dangling pointers, out-of-bounds violations for array indexing, memory leaks, and buffer overflows do not set off exception-handling logic. Rather causing disastrous runtime behavior, such as over consumption of system resources, unintentional read/write to memory, or overall system crashes [TTT12]. Despite the faster run-time, the procedural design of C/EC lacks object-oriented programming constructs. Principles such as encapsulation, polymorphism, and constructs, are not present for C/EC. For large-scale embedded projects, involving multiple concurrent communication stacks, or other extensions, this can cause rapid degradation of projects/codebases into unmaintainable messes [TTT13]. Requiring developers

to maintain strict observation and discipline to prevent any issues or deterioration.

In the context of the BEACON project, C/EC is an excellent programming language for developing the firmware of both microcontrollers. The ability to directly manipulate hardware registers and memory addresses without losing the high-level readability, or requiring on-chip interpreters, makes EC the logical choice for firmware design. Furthermore, the compiled nature of C/EC allows for unparalleled execution speed and power efficiency, which the handlebar system will greatly benefit from. Unfortunately, extra care will have to be taken to avoid as many memory-leakage issues as possible. Since rider safety is paramount, system crashes, buffer overflows, or other issues that could affect system performance must be avoided at all costs. To ensure proper system integrity, regular code and performance reviews will be run during testing of the BEACON project.

3.8.1.2 C++/Embedded C++

C++ is an object-oriented superset of C that was created to combine the speed and efficiency of the C programming language with object-oriented programming paradigms. Similar to C with its embedded systems extension EC, C++ contains its own extension appropriately named “Embedded C++” (abbreviated to “EC++”). The core guiding principle of EC++ is the “Zero-overhead principle”, dictating that software abstractions should not suffer from runtime or storage penalties compared to hand-written procedural code [TTT14]. Like with C, C++ is compiled directly into machine code, without requiring an on-chip interpreter or a virtual machine, allowing developers to interact with the physical hardware registers through raw pointer manipulation. Unlike C, however, EC++ also includes object-oriented functionality, allowing for extra functionality and organization for larger-scale projects.

The driving philosophy behind EC++ (and C++ as a whole) is the zero-cost abstraction technique. Zero-cost abstraction aims to completely remove the performance overhead associated with using abstractions by utilizing compile-time optimizations that eliminate runtime-introduced overhead [TTT14]. EC++ offers large support for the zero-cost abstraction concept, granting developers access to object-oriented programming in environments with limited resources, such as embedded systems. Now, through this methodology, embedded developers using EC++ gain access to a toolkit that elevates and enhances embedded firmware. For example, when writing driver abstractions through standard EC, developers need to use function pointers or void* casts to abstract the physical hardware bus. With EC++, templates generate compile-time polymorphism, in which the compiler creates unique, optimized variants of the driver class at build time [TTT15]. Bypassing the entire function call overhead and turning structural abstraction into direct sequential assembly instructions.

Similar to C/EC, C++/EC++ can directly access and manipulate hardware address registers. Unlike other “high-level” languages, EC++ does not require a virtual machine or on-chip interpreter to work with the MCU hardware directly. Instead, a program written in EC++ can directly interact with the hardware register addresses, significantly reducing run-time performance penalties. In comparison to other programming languages, like Python, an MCU’s CPU can execute a line of code in 1-2 clock cycles (depending on the CPU). When implemented in a system similar to the BEACON project, where quick-time analysis is paramount, the direct translation to hardware code can reduce response jitter for sensors or eliminate display lag.

Despite the benefits, C++/EC++ are often considered not suitable for embedded systems. EC++ can introduce significant architectural risks, and if not managed properly, software overhead, despite the zero-cost abstraction methodology. When implementing high-level features, the compiler optimizes execution speed by trading away code compactness. For a resource-limited environment, if used carelessly, can result in excess flash memory costs, appropriately named “code bloat” [TTT14]. This can lead to performance issues from increased cache misses and longer load times [TTT14]. Furthermore, EC++ contains exception handling, like most high-level OOP languages. If a runtime error is encountered through exception handling, the EC++ runtime environment will begin unwinding the stack, searching backwards through the execution memory to find the matching catch block [TTT14]. Lastly, with the heavy emphasis that C++/EC++ puts on dynamic object creation “behind the scenes” [TTT14], the limited SRAM of the average MCU can succumb to memory fragmentation [TTT14].

When considering C++/EC++ for the BEACON project, it would at first seem like the perfect choice. By using EC++, developers gain access to OOP paradigms along with direct access to high-speed hardware control and manipulation. Seemingly, the best of both worlds. However, when considering the potential risks, EC++ requires more care to ensure that the limited resources of the MCU are not wasted on dynamically-allocated objects. While for a larger-scale project it would be worth the risk, the BEACON project is not intended to reach the scale necessary to justify the risks.

3.8.1.3 Rust

Rust is a multi-paradigm, statically typed, systems programming language designed for performance, reliability, and memory safety. Rust is capable of achieving the compilation speeds of C/C++ and also eliminates security vulnerabilities, such as null pointer exceptions and buffer overflows. Rust has built-in safety features that are designed specifically for handling the memory safety issues common in both C and C++. Furthermore, similar to both C and C++, Rust compiles directly to native machine code without the use of an interpreter or virtual machine. Adopted by major tech companies and endorsed

by safety agencies. Rust has rapidly become an industry standard for embedded systems.

The two main reasons for Rust to be implemented in embedded systems design is the safety and reliability provided by the language [TTT16]. Rust contains multiple features for error-handling, such as the “panic” macro, that stops execution when a Rust program encounters an unrecoverable error [TTT17]. Additionally, the Rust programming language implements an “ownership model” that guarantees compiler-memory safety. Through the ownership model, the compiler tracks the lifecycle of every variable, pointer, or value [TTT17]. During compile time, ownership is checked, and if any of the rules are violated, the program will not compile. Another system to ensure memory safety is the Lifetime mechanism. Lifetimes are a kind of generic typing that ensures references are valid for as long as they’re needed by a developer [TTT17]. Every reference in Rust has a scope for which it’s valid, referred to as the lifetime of the reference. Through the use of reference lifetimes, the Rust compiler is able to ensure that they are still valid, avoiding memory related errors such as dangling references/pointers [TTT17].

Despite the safety benefits, Rust introduces unique technical challenges, such as compilation time and hardware mismatch. One of Rust’s greatest safety features, the ownership model, actually causes significant hardware conflicts. Because Rust assumes a structured and sequential flow of execution, where references have a clear lifetime, the asynchronous nature of microcontrollers can cause a false positive on Rust’s memory safety features. When an interrupt service routine (ISR) halts the main program thread, the Rust safety features will kick in and recognize this behavior as a violation of the “one mutable reference at a time” ownership rule [TTT17]. Similar to C++, Rust projects can experience code bloating. For Rust specifically, experienced engineers have considered it to have a steep learning curve. Which makes code bloating more likely, especially when considering Rust’s type system, compile-time checks, and formatting macros that can lead to larger binary footprints [TTT18].

In the context of the BEACON project, while Rust sounds like a very promising candidate for firmware development, similar to C++ the downsides outweigh the benefits. The steep learning curve mixed with the memory-safety technology native to Rust introduces a unique challenge that can cause significant development roadblocks. Furthermore, Rust’s type system and heavy reliance on generics can lead to larger code bloating. Which would present a constant issue on resource-limited microcontrollers. For these reasons among many, Rust has been hesitantly accepted by other embedded developers and is also why the BEACON project will implement a different programming language for firmware development.

3.8.2 Programing Language Selection

3.8.2.1 Programming Languages Comparison Table

	C/EmbeddedC	C++/EmbeddedC++	Rust
Primary Purpose	MCU programming	OOP alternative to C	Memory-Security focused alternative to C
Features	Pointers and direct memory addressing, little to no overhead cost.	The same features as C with additional OOP paradigms.	Ownership tracking prevents compile-time memory issues such as dangling references
Memory Management	Manual management by developer	Scope-based destructors	Ownership tracking
Challenges	No safety nets and weak compile-time verification	Hidden code bloat, error handling can cause issues with deterministic real-time execution	Interrupt handling mismatch from ownership and steep learning curve
Advantages	Portable, minimal overhead cost, and 1:1 code to physical assembly mapping	Zero-cost abstractions and reduced compile-time compilation	Memory safety through ownership and built-in concurrency protection

The BEACON project contains many unique peripherals, hardware, and components that all use different communication protocols to work together. Because of this, when developing the firmware for the MCUs, each programming language needs to be carefully considered. After going over the pros, cons, and the general description of each of the above languages, it would seem that C/EC would be the best fit for the BEACON project. EC provides minimal computational overhead along with no hidden abstractions causing code bloat, or safety features that cause hardware conflictions. However, the BEACON project is ongoing. As progress is made, there may be cases where either EC++ or Rust is needed. At which point, they will be carefully considered again, and implemented accordingly.

3.8.3 Algorithms

The BEACON project has many different hardware components, each with differences in operation. Because of this, several different algorithms will be employed to ensure proper functionality of the system as a whole. For example, the digital signal processing associated with the radar sensor, or the data structures involved with the communication between peripherals and MCU. There are many different algorithmic solutions to each task that the BEACON project must fulfill in terms of firmware/software. Therefore, they each need to be considered and compared to one another. Furthermore, it is inevitable that there will be design trade offs during production. Another goal of considering each solution is determining what trade off is more beneficial than harmful.

One of the primary features for the BEACON project is object detection using radar. In general, radar sensors emit electromagnetic waves that bounce off of objects. Through the use of electromagnetic waves, radar sensors are highly reliable and can be used in extreme weather or environments with dust or drastic temperature changes.

In the case of the BEACON project, our radar is expected to detect fast moving vehicles. This means that the radar module will be using the doppler effect to get data of the environment around it [TTT19]. However, the data returned still needs to be processed and filtered to get any real value out of it. Signal processing and filtering has many different algorithmic approaches, such as the moving average filter, the exponential moving average filter, or the wavelet transform technique.

For the BEACON project, the radar detection system must have a fast response time. Enough for the rider to recognize and respond accordingly. To achieve the signal processing algorithms need to provide low latency and high accuracy.

3.8.3.1 The Moving Average Filter

The moving average filter (MAF) is one of the most common digital signal processing filters [TTT20]. Of the digital signal processing techniques mentioned, it is the easiest to understand, and the easiest to implement [TTT20]. By smoothing data through an average set number of consecutive samples, the MAF is able to reduce random noise and still retain a sharp step response [TTT20]. The smoothing process of the MAF decreases the amplitude of random background noise, but also reduces the sharpness of edges [TTT20].

However, despite how perfect the MAF may seem, it still has its drawbacks. For example, the MAF cannot separate one band of frequencies from another [TTT20]. Furthermore, the MAF allows a significant amount of high frequency noise to leak through. Overall, the MAF is best used for smoothing frequencies but is an exceptionally bad low-pass filter [TTT20]. Considering the radar system employed by the BEACON project requires the possible detection of multiple objects, the MAF will likely not be able to fulfill that requirement.

3.8.3.2 The Exponential Moving Average Filter

Unlike the MAF, the exponential moving average filter (EMAF) is a first-order infinite impulse response (IIR) filter [TTT21]. IIR filters, like the EMAF, are useful due to their capability to bypass a longer convolution [TTT21]. When looking at the time domain performance, the EMAF excels at reducing latency. By assigning the smoothing factor the largest weight to the most recent data point, the EMAF can track sudden shifts in signals much faster than a MAF [TTT21].

For the frequency domain, the EMAF performs like a single pole low-pass filter [TTT21]. However, the EMAF still possesses the gradual roll-off seen in first order filters, like the MAF [TTT21]. In general, EMAF are exceptionally efficient. However, the trade off for this efficiency is a drop in performance and flexibility in comparison to other digital filters [TTT21].

The EMAF provides an exceptionally fast computation time in comparison to the MAF in addition to compensating for the MAF's significant high frequency leakage. Unfortunately, it still suffers from the roll-off characteristic of typical first order filters. In consideration of the BEACON project, the EMAF is a promising candidate for digital signal processing and noise reduction. Despite the fact that the slow roll-off offers considerable issues in noisy environments, like a busy street.

3.8.3.3 The Wavelet Transform

The wavelet transform (WT) provides an alternative to the standard Short-Time Fourier Transform (STFT) [TTT22]. The WT uses short windows at high frequencies and long windows at short frequencies, which is referred to as "constant relative bandwidth frequency analysis" [TTT22]. There are two main approaches to the WT technique, each with their own computational cost. For the purposes of this project, the discrete wavelet transform (DWT) will be the primary focus. As the computational cost for the DWT has been reduced to strictly $O(N)$, where N is the number of data points. Making the DWT faster than using the Fast Fourier Transform (FFT) [TTT22].

As mentioned previously, the DWT stretches and compresses based on the frequency band being analyzed. For higher frequencies, the wavelet is compressed, allowing the filter to capture sudden, sharp spikes with exceptional accuracy [TTT22]. This particular case would be very beneficial for the BEACON project. For riders on busy roads, a sudden sharp spike from a car rapidly approaching the radar sensor would not be a rare event. For lower frequencies, the wavelet is expanded, allowing the filter to measure slow-moving baseline frequencies [22]. Unlike the previously mentioned filters, the wavelets used in the DWT technique can isolate specific features of a signal [22]. Making them mathematically optimal for distinguishing signals from background noise.

Similar to the EMAF, the DWT implementation of the WT technique seems to be a perfect fit for the BEACON project. However, one major issue for the WT technique comes from the implementation complexity. Unlike the previously mentioned filters, the WT provides considerable challenges in terms of algorithmic complexity. Requiring the application of a wavelet coefficient matrix along with “smoothing” vectors. The algorithm described is referred to as the “pyramidal algorithm” [23]. When regarding the BEACON project, the WT technique is very promising and would most likely provide the exact solution to digital signal processing and noise filtering. However, the implementation challenges that it provides may be too much to handle in the time-frame allowed for the project’s time frame.

3.8.4 Algorithm Comparison

3.8.4.1 Algorithm Comparison Table

Algorithm	Latency	Accuracy	Computational Complexity	Implementation Complexity
Moving Average Filter	Very Low	Circumstantial (low-moderate)	Low	Simple
Exponential Moving Average Filter	Very Low	Circumstantial (moderate-high)	Low	Simple
Wavelet Transform Technique	High	Very High	Very High	Very Complex

4.0 Standards and Design Constraints

4.1 Industry Standards

ISO 6742-1:2023 – Cycles: Lighting and Light Signalling Devices

The ISO 6742-1:2023 standard establishes the performance and safety requirements for lighting and light signalling devices used on bicycles intended for operation on public roads [SSS26]. The standard specifies the required functions, photometric performance, safety requirements, and testing methods for bicycle lighting systems to ensure cyclists remain visible to surrounding road users under a wide range of operating conditions [SSS26]. Unlike general lighting recommendations, ISO 6742-1 focuses specifically on bicycle applications by defining requirements for rear lamps, stop lamps, direction indicators, daytime

running lamps, and other signalling devices that improve communication between cyclists and motorists [SSS26]. By establishing standardized performance requirements, the standard promotes improved cyclist visibility, reduces the likelihood of collisions, and provides manufacturers with consistent testing methods for evaluating lighting performance [SSS26]. These requirements are highly applicable to the B.E.A.C.O.N system since its primary objective is to increase cyclist visibility while clearly communicating rider intentions through brake lights, turn signals, and vehicle detection alerts.

The primary objective of ISO 6742-1 is ensuring that bicycle lighting devices remain visible under a variety of environmental conditions while minimizing ambiguity for surrounding motorists [SSS26]. The standard defines several categories of lighting devices, including rear lamps that indicate the presence of the bicycle, stop lamps that communicate braking or rapid deceleration, and direction indicators that notify nearby traffic of an intended turn [SSS26]. In addition to defining these functions, the standard specifies photometric performance requirements that ensure lighting devices produce sufficient luminous intensity while maintaining appropriate beam distribution and visibility angles. These performance requirements are verified through standardized testing procedures that evaluate the lighting system under repeatable laboratory conditions, ensuring that products from different manufacturers provide consistent performance [SSS26].

The visibility requirements outlined within ISO 6742-1 significantly influenced the development of the B.E.A.C.O.N system. Since one of the primary goals of the project is improving cyclist conspicuity, high-intensity L.E.Ds were selected to maximize visibility during both daytime and nighttime riding [SSS26]. Research has shown that increasing the conspicuity of cyclists through active lighting significantly improves the ability of motorists to detect cyclists and respond appropriately under varying traffic conditions [SSS27]. Particular attention was given to the rear warning module, where red L.E.Ds were selected to clearly distinguish the bicycle from surrounding traffic while maintaining low power consumption [SSS26]. The use of dedicated brake light functionality further improves rider communication by providing a clear visual indication whenever rapid deceleration occurs. This additional signalling capability allows following motorists to react more quickly than would be possible using only a conventional rear reflector or continuously illuminated tail light [SSS26], [SSS27].

Direction indication is another important feature addressed by ISO 6742-1. The standard identifies direction indicators as lighting devices used to communicate a cyclist's intention to change direction, helping surrounding road users anticipate upcoming manoeuvres [SSS26]. For bicycles operating in urban traffic, this communication is particularly important because cyclists frequently share road space with larger vehicles whose drivers may have limited visibility. Studies investigating cyclist visibility have demonstrated that clear visual cues significantly improve driver recognition of cyclist intentions, particularly at

intersections and during lane changes [SSS27]. Within the B.E.A.C.O.N project, this requirement directly influenced the incorporation of dedicated left and right turn signals. Rather than relying solely on traditional hand signals, which may be difficult to observe during poor weather, nighttime operation, or heavy traffic, electronically controlled turn indicators provide a highly visible and consistent indication of rider intent. This improves communication between the cyclist and surrounding motorists while reducing the likelihood of misunderstandings during lane changes and turning manoeuvres [SSS26], [SSS27].

Another important consideration within ISO 6742-1 is the evaluation of lighting performance through standardized photometric testing. Rather than simply requiring that a light emit illumination, the standard verifies that lighting devices maintain sufficient brightness, beam distribution, and visibility across multiple viewing angles [SSS26]. These requirements are particularly important because bicycles may be observed from directly behind, from adjacent traffic lanes, or while approaching an intersection. Consequently, the B.E.A.C.O.N lighting modules were designed to maximize visibility across a wide range of observation angles by carefully selecting L.E.D placement and mounting locations. Positioning the lighting assemblies near the rear of the bicycle ensures that the brake lights and turn signals remain clearly visible without obstruction from the rider, cargo, or bicycle frame. Designing the lighting system around these photometric requirements improves the likelihood that the signalling devices remain effective under a variety of real-world riding conditions [SSS26].

Compliance with ISO 6742-1 also supports future product development beyond the prototype stage. Although the current B.E.A.C.O.N system is intended as a proof-of-concept design, developing the lighting system around internationally recognized bicycle lighting standards simplifies future testing, certification, and commercialization [SSS26]. Designing according to established lighting performance requirements reduces the likelihood that significant hardware redesigns would be required before large-scale production. Furthermore, aligning the project with recognized international standards demonstrates that engineering decisions were based on accepted industry practices rather than arbitrary design choices, improving the credibility and potential manufacturability of the final design [SSS26].

By following the principles established within ISO 6742-1:2023, the B.E.A.C.O.N project provides improved cyclist visibility, enhanced communication of rider intentions, and greater overall road safety [SSS26]. The standard directly influenced the selection of lighting components, signal functions, mounting locations, and overall system architecture. Applying these internationally recognized requirements ensures that the final design not only satisfies the functional objectives of the project but also aligns with current engineering practices for bicycle lighting and signalling systems [SSS26]. Combined with published research on cyclist conspicuity and driver perception, these design decisions provide a strong engineering basis for improving cyclist safety in real-world traffic environments [SSS27].

IEC 60529 – Degrees of Protection (IP Code)

The International Electrotechnical Commission (IEC) 60529 standard establishes the Ingress Protection (IP) Code, which classifies the degree of protection provided by electrical enclosures against the intrusion of solid foreign objects and water [SSS28]. The standard provides a uniform method for evaluating enclosure performance by assigning a two-digit IP rating, where the first digit specifies protection against solid particles such as dust, while the second digit specifies protection against water ingress under various exposure conditions. By standardizing these classifications, IEC 60529 allows manufacturers and consumers to compare products based on their ability to withstand environmental conditions and ensures that electrical equipment can operate safely and reliably within its intended application [SSS28].

For outdoor electronic devices such as the B.E.A.C.O.N bicycle safety system, ingress protection is a critical design consideration. Unlike indoor electronics, bicycle-mounted devices are routinely exposed to rain, road spray, dust, mud, and debris generated during normal riding conditions. Environmental contaminants entering the enclosure could damage sensitive electronic components, corrode electrical connections, or create short circuits that compromise system functionality. Since the primary purpose of the B.E.A.C.O.N system is to improve cyclist safety through continuous operation of its lighting and vehicle detection features, maintaining reliable operation under adverse environmental conditions is essential. Consequently, the enclosure was designed with the objective of achieving a level of protection consistent with the recommendations of IEC 60529 for outdoor electronic equipment [SSS28], [SSS29].

One of the most important aspects of IEC 60529 is the protection against solid foreign objects. The first digit of the IP rating ranges from 0 to 6, where higher values indicate greater protection against the ingress of dust and other particles [SSS28]. Bicycle operation frequently exposes electronic devices to airborne dirt, gravel dust, road debris, and other contaminants that can accumulate over time. If these particles enter the enclosure, they may interfere with switches, connectors, charging ports, or printed circuit boards, reducing reliability and increasing maintenance requirements. To minimize these risks, the B.E.A.C.O.N enclosure incorporates tightly fitted enclosure interfaces and carefully positioned openings that limit the possibility of dust entering the device while still allowing practical assembly and servicing. These design considerations improve long-term durability while reducing the likelihood of premature component failure caused by environmental contamination.

Protection against water ingress represents the second major consideration defined within IEC 60529. The second digit of the IP rating specifies the enclosure's ability to resist water under conditions ranging from light dripping to

high-pressure water jets and temporary immersion [SSS28]. Since bicycles are frequently operated during rainfall or on wet road surfaces, water resistance is essential for maintaining safe and reliable operation. Moisture entering the enclosure could cause corrosion of exposed conductors, electrical shorts, sensor malfunction, or battery damage, any of which could disable critical safety functions such as brake lights, turn signals, or rear vehicle detection. For this reason, the enclosure design incorporates sealing features around enclosure joints, external fasteners, and cable entry locations. Additionally, consideration was given to protecting the charging interface through the use of sealing covers or recessed connector placement, reducing the likelihood of water reaching sensitive electrical contacts [SSS28], [SSS29].

Compliance with IEC 60529 also influenced several mechanical design decisions beyond simple sealing techniques. Achieving higher levels of ingress protection often requires careful consideration of enclosure materials, gasket selection, manufacturing tolerances, and assembly methods. Small gaps created by poor manufacturing tolerances or improper assembly can significantly reduce the effectiveness of environmental seals, even when high-quality materials are used. Therefore, the enclosure geometry for the B.E.A.C.O.N system was developed to minimize potential leak paths while remaining practical to manufacture using common fabrication techniques. These considerations improve the robustness of the overall design while avoiding unnecessary manufacturing complexity or excessive production costs.

Designing for a higher IP rating also introduces engineering trade-offs that must be carefully balanced. Increasing enclosure protection often results in more complex manufacturing processes, additional sealing components, and reduced accessibility for maintenance or battery replacement. Completely sealing an enclosure may also reduce airflow, limiting passive cooling and increasing internal operating temperatures during extended use. Consequently, the design process required balancing environmental protection with thermal management, manufacturability, serviceability, and overall product cost. Rather than maximizing the IP rating regardless of consequence, the objective was to select an appropriate level of protection that satisfies the intended operating environment while maintaining practical manufacturing and maintenance considerations [SSS28].

Although the current B.E.A.C.O.N system is being developed as a prototype, designing in accordance with IEC 60529 provides significant advantages for future commercialization. Products intended for consumer use are expected to demonstrate reliable performance under realistic environmental conditions, and compliance with internationally recognized ingress protection standards provides confidence that the device can withstand routine outdoor operation. Designing around these requirements also reduces the likelihood of significant enclosure redesigns during future product refinement, simplifying certification and increasing the commercial viability of the system.

By following the principles established within IEC 60529, the B.E.A.C.O.N project improves the durability, reliability, and long-term performance of its electronic systems when operating in outdoor environments. The standard directly influenced enclosure design, sealing methods, material selection, connector placement, and overall mechanical architecture. Applying these internationally recognized ingress protection requirements ensures that the final design not only satisfies the functional objectives of the project but also aligns with accepted engineering practices for outdoor electronic products intended for daily use [SSS28], [SSS29].

IPC-2221 – Generic Standard on Printed Board Design

The IPC-2221 standard establishes the general requirements for the design of printed circuit boards (PCBs) used in electronic systems. Published by the Association Connecting Electronics Industries (IPC), this standard provides comprehensive design guidelines covering conductor spacing, trace width, current-carrying capacity, component placement, thermal management, electrical isolation, and overall board reliability [SSS31]. Rather than prescribing a single PCB layout, IPC-2221 serves as a foundational engineering standard that assists designers in producing safe, reliable, and manufacturable electronic circuits. By following these recommendations, engineers can reduce manufacturing defects, improve electrical performance, and increase the long-term reliability of electronic products operating in a variety of environmental conditions.

The B.E.A.C.O.N bicycle safety system relies heavily on electronic hardware to perform its primary functions, including controlling the brake lights, turn signals, vehicle detection system, battery management circuitry, and user interface. These subsystems are integrated onto one or more printed circuit boards that distribute electrical power and communication signals throughout the device. Since the PCB serves as the backbone of the electronic system, careful consideration of PCB design practices was necessary to ensure reliable operation. Applying the design recommendations outlined in IPC-2221 allows the B.E.A.C.O.N system to minimize electrical failures while improving manufacturability and simplifying future revisions [SSS30].

One of the most significant design considerations addressed by IPC-2221 is conductor sizing and current-carrying capacity. Copper traces on a PCB must be wide enough to safely carry electrical current without excessive heating or voltage drop. Undersized traces can generate excessive temperatures during operation, potentially damaging the PCB substrate, degrading solder joints, or causing premature component failure. Conversely, excessively wide traces unnecessarily increase board size and manufacturing cost. Since the B.E.A.C.O.N system incorporates multiple high-intensity L.E.Ds that require relatively high drive currents compared to logic circuitry, appropriate trace widths were considered during PCB development to ensure reliable power delivery while

maintaining a compact circuit board layout. Power traces supplying the lighting system were designed separately from low-current signal traces used by sensors and microcontrollers to reduce electrical interference and improve overall circuit stability [SSS30], [SSS31].

Component spacing is another critical design requirement identified within IPC-2221. Proper spacing between electrical conductors minimizes the possibility of unintended electrical arcing, solder bridging, and manufacturing defects while also improving accessibility during assembly and inspection [SSS30]. Because the B.E.A.C.O.N. system combines power electronics, battery charging circuitry, microcontroller components, sensors, and communication interfaces within a relatively small enclosure, maintaining appropriate spacing between these circuits was an important design consideration. Adequate separation between high-current L.E.D drivers and low-voltage digital circuitry reduces electrical noise and improves signal integrity, ensuring that communication between sensors and processing hardware remains reliable during operation.

Thermal management also represents a major consideration addressed by IPC-2221. Electronic components naturally generate heat during operation, particularly power regulators, L.E.D drivers, charging circuits, and high-brightness L.E.Ds. If this heat is not effectively dissipated, elevated temperatures may reduce component lifespan, decrease L.E.D efficiency, and increase the likelihood of electronic failure. IPC-2221 provides recommendations regarding copper plane utilization, thermal relief patterns, component placement, and board layout to assist designers in managing heat generation [SSS30]. These recommendations directly influenced the arrangement of high-power components within the B.E.A.C.O.N. PCB. Heat-generating components were positioned to maximize thermal dissipation while minimizing heat transfer to temperature-sensitive devices such as the microcontroller and battery management circuitry. Maintaining acceptable operating temperatures contributes to improved system reliability and extends the expected service life of the electronics.

Another important aspect of IPC-2221 is manufacturability. A well-designed PCB should not only function correctly but should also be practical to manufacture, assemble, inspect, and repair. The standard provides recommendations regarding pad geometry, via placement, solder mask clearances, and component orientation that simplify automated assembly processes while reducing the likelihood of manufacturing defects [SSS30]. Although the B.E.A.C.O.N project is currently being developed as a prototype, these manufacturability considerations remain important. Designing the PCB using accepted industry practices improves the consistency of prototype fabrication while reducing the likelihood of costly redesigns if the project progresses toward commercial production. Furthermore, standardized PCB design practices make future troubleshooting and component replacement significantly easier.

Electrical reliability is another area significantly influenced by IPC-2221. Since the B.E.A.C.O.N. system operates on a bicycle that is routinely exposed to vibration, impacts, and changing environmental conditions, the PCB must maintain reliable electrical connections throughout normal operation. Improper component placement or insufficient mechanical support can lead to solder joint fatigue and intermittent electrical failures over time. Following the recommendations outlined within IPC-2221 improves mechanical robustness by encouraging balanced component placement, appropriate mounting locations, and adequate structural support throughout the circuit board. These design practices increase the long-term durability of the electronics while reducing maintenance requirements for the end user.

Compliance with IPC-2221 also benefits future product development by providing a standardized design methodology recognized throughout the electronics industry. As additional features are incorporated into future versions of the B.E.A.C.O.N. system, such as wireless communication, expanded sensing capabilities, or advanced battery management, following established PCB design standards simplifies future modifications without compromising reliability. Designing according to IPC recommendations also improves compatibility with commercial PCB manufacturers and contract assembly services, reducing manufacturing risk during future product development.

By applying the design principles established within IPC-2221, the B.E.A.C.O.N project improves the reliability, manufacturability, thermal performance, and electrical integrity of its electronic systems. The standard directly influenced conductor sizing, component placement, thermal management, PCB layout, and overall hardware architecture. Applying these industry-recognized design practices ensures that the electronic hardware supporting the bicycle safety system is both technically sound and suitable for future development beyond the prototype stage [SSS30], [SSS31].

4.2 Design Constraints

4.2.1 Electrical Constraints

One of the most significant design constraints for the B.E.A.C.O.N system is that of its electrical system, as it integrates multiple electronic subsystems including high-intensity L.E.D brake lights, turn signals, a rear vehicle detection sensor, microcontrollers, and supporting power management circuitry. Maintaining stability and efficiency in these components is necessary for reliable operation, a strict requirement for a safety system. Unlike stationary electronic devices that operate from a continuous power source, B.E.A.C.O.N operates entirely from rechargeable lithium-ion batteries, meaning every component must be designed with power efficiency in mind. Excessive power consumption reduces battery life, while poor voltage regulation or electrical noise can negatively impact system performance. Consequently, electrical design considerations influenced nearly

every aspect of the project, from component selection to power distribution and PCB layout.

Although a larger battery could significantly increase operating time, doing so would also increase the overall size, weight, and cost of the system. Conversely, selecting a battery with insufficient capacity would require frequent charging, reducing practicality for everyday use. Finding an appropriate balance between battery capacity, physical size, and operating time therefore became an important constraint throughout system design. Likewise the selection of power efficient components, particularly the use of high-efficiency L.E.Ds and, helps maximize runtime without unnecessarily increasing battery size.

Voltage regulation is another relevant constraint. Individual components within B.E.A.C.O.N operate at different supply voltages, requiring stable voltage conversion from the battery pack to the appropriate operating levels. The selected switching regulator provides efficient voltage conversion while minimizing energy loss that would otherwise occur with linear regulation. High conversion efficiency is especially important for battery-powered devices because wasted electrical energy is converted directly into heat, reducing both runtime and system efficiency. Stable voltage regulation also ensures that the microcontroller, detection sensor, and communication interfaces receive a consistent supply voltage regardless of battery charge level, allowing the system to operate reliably throughout the battery's discharge cycle. Electrical noise and signal integrity also are taken into account. B.E.A.C.O.N contains both high-current and low-current circuitry operating simultaneously on the same printed circuit board. The L.E.D drivers and switching regulator generate rapidly changing currents that can introduce electrical noise into nearby signal traces if proper PCB layout techniques are not followed. At the same time, the rear vehicle detection sensor and microcontroller rely on clean electrical signals for accurate object detection and reliable data processing. Excessive electrical noise could lead to unstable sensor readings and errors.. To account for these risks, it is important to separate power and signal circuitry, follow proper grounding practices, and appropriate PCB trace routing as a part of power management.

5.0 Acknowledgement of ChatGPT and Other Similar Platforms

5.1 Introduction

Throughout the history of engineering, many different tools and techniques have been developed, not only to aid in its physical challenges, but also to assist in the practice of collecting and synthesizing information. In this way, it is important to recognize that any project not only relies on the technical knowledge and experience of its team members, but arguably more importantly, their ability to learn new information and adapt to the challenges presented to them. This is why

methods such as brainstorming, research, and prototyping, have been an especially important pillar of the engineering process.

Traditionally, gathering information for a project has required engineers to turn towards places like universities, libraries, or even the industry for past work and information. Now, in the last 3 decades, the internet has enabled unfettered access to an enormous range of resources at our finger tips. Modern tools, like the search engine, have helped to streamline the process of collecting the relevant knowledge needed to develop new solutions.

Despite its immense impact, the internet has still required its users to not only have knowledge of what they are searching for, but to also follow the selective process of actively locating the specific resources they are looking for. Even more so, it had remained necessary for engineers to read material and retain the knowledge needed for their own projects.

That being said, once again, the practice has begun to change; this time at an ever increasing rate with the development of A.I. Specifically, the large language models we have today like: Gemini, Copilot, Claude, DeepSeek, Perplexity, and ChatGTP, which have created platforms to allow for anyone to not only access information, but to have it synthesized for them. This marks a distinct shift in the previous methodology, where now people can not only be provided the location of any source of information, but can be told what they should even be looking for in the first place.

The models being discussed are a form of deep learning neural networks that use transformer architecture. Effectively, they operate as a statistical predictor that works on sequences of words. These models break down the large amount of information they are trained on into tokens for words and characters, allowing the transformer network to calculate relationships between tokens and map them into a vector space, forming associations that help it to produce the models results based on user prompting [SSS6].

Being language models, they appear as something that can be communicated and reasoned with, and the presentation of their results makes it common for users to perceive these tools as a reliable source of information. However, since these models are associative, they are only mimicking human behavior, requiring more nuance to ensure credibility. Ultimately questions arise on the validity of their use, especially in professional and academic settings.

5.2 Application

Large language A.I models can be used for a wide variety of uses, but most practically they operate by turning user prompts into generated outputs, being able to answer and solve questions, provide ideas, create code, generate images, and much more. Since their scope is so wide, this report will specifically

focus on how these models can be best used within an engineering context, and will exclude general conversations on ethics. It is possible to use these tools in brainstorming, research, and prototyping.

One of the strongest benefits of A.I is its ability to summarize large amounts of information that would otherwise require reviewing multiple sources. A user is able to explain what they are trying to do, and instead of having to move from resource to resource to try and find it, an A.I can give information in a condensed summary. This helps to save time and narrow down possible options to focus attention on.

In addition, the other major feature of the tool is its conversational structure. Multiple prompts can be saved and chained together, allowing for the model to respond to user preferences. This allows for prompts to compound on one another and the model to change how it presents information to best suit user preferences. For example, if a response was hard to understand, the user could request for it to be worded in a different way, or to use different formatting. This responsiveness can help to make the learning process more effective by allowing for more flexibility.

Since the results of the model are based on user prompts, it is necessary for users to apply specific methodology when interacting with these models. Specifically, when creating a prompt, the quality of the results will depend on how specific it is, preventing the model from straying too far from the desired answer. This idea leads into one of the biggest issues for A.I, the illusion of credibility. Since A.I is generative, there is no guarantee about the validity of its response, as the model operates under a false confidence. Recognizing this limitation is important, as verifying the sources it uses and investigating them is necessary.

The other important problem to recognize when using A.I is the risk of losing expertise and found understanding in a subject. Since these models are able to effectively provide simple and general explanations for almost any subject, it is especially important in a research setting not to rely on it for the grounds of your personal understanding, else forming the needed expertise for a project will likely prove difficult. The information they provide is not comprehensive, as it is curated only to the prompts you provide, and it does not challenge you in a way that facilitates growth or a critical understanding of the topic. When looking for unique and novel ideas, it is necessary to build up your own expertise to solve original problems [SSS7].

5.3 Project Examples

5.3.1

I used Chat GPT to help me structure and find sources for my batteries research. ChatGPT provided me with a few sources to look into as well as

provide a structure to follow to help the research be organized and formal [BBB15]. Another example is the start of my antenna research where I asked chatgpt to compile a list of antennas and explain why I would use them for this project [BBB16]. Using AI in this way helped me lay the foundation to further the research on my own and set me on the right course.

6.0 System Hardware Design

6.1 Power

The power subsystem within B.E.A.C.O.N is responsible for supplying stable electrical power to every other component and subsystem in the design. Since all on board electronics rely on the power subsystem to operate, with its single power source, the design of this subsystem directly impacts the overall reliability, operating time, and performance of the project. This power system was designed to convert the battery voltage into the supply rails of the microcontroller, L.E.D driver circuitry, sensors, and supporting peripherals while minimizing power losses to improve total battery life.

To meet these requirements, B.E.A.C.O.N is powered with a rechargeable two-cell lithium-ion battery pack using Samsung INR18650-35E cells. These cells utilize NMC chemistry, notable for its high energy density, lightweight construction, long cycle life, and improved thermal stability compared to alternative lithium-ion chemistries. These characteristics make NMC cells particularly well suited for portable electronic devices where both runtime and overall system weight are important design considerations. Each Samsung cell provides a nominal voltage of 3.6 V, a nominal capacity of 3350 mAh, and is capable of supplying up to 8 A of continuous discharge current. By connecting two cells in series, the battery pack provides a nominal output voltage of approximately 7.2 V with a fully charged voltage of 8.4 V and a fully discharged operating voltage of approximately 6.0 V. Together, these characteristics compile to approximately 24 Wh of energy.

After leaving the battery packs, the input voltage is routed to the power management circuitry where it is distributed to two independent buck converters. These converters generate the regulated 5 V and 3.3 V power rails required by the system. Separating the power conversion into two dedicated rails allows each subsystem to receive a stable supply voltage while improving efficiency and reducing electrical noise between high-current and low-power circuitry.

6.1 3.3V Buck Regulator

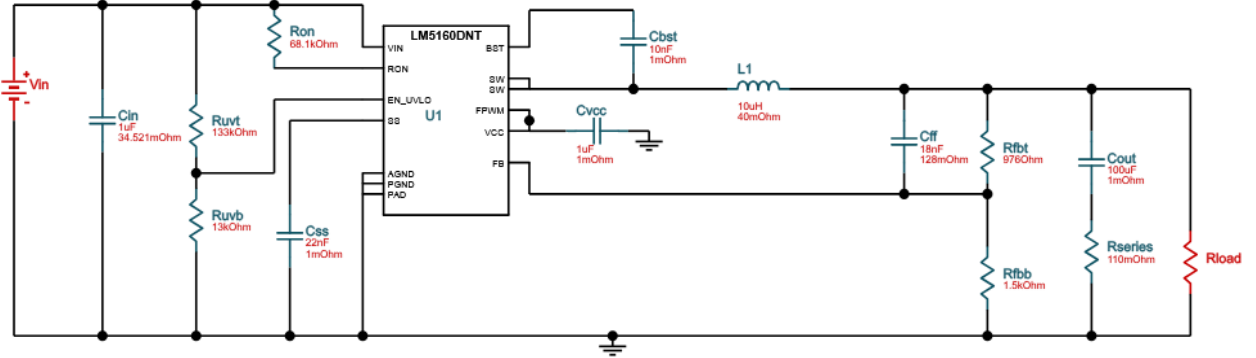


Figure 6.1.1 3.3V Buck Regulator

The 3.3 V regulator uses the Texas Instruments LM5160 synchronous buck regulator, as shown in figure above. This regulator was selected using the Texas Instruments WEBENCH Power Designer for its low ripple current, while also maintaining a high conversion efficiency, BOM price. These specifications are well suited for the compact battery-powered design of the B.E.A.C.O.N. system. Since it is powered from a two-cell lithium-ion battery pack, with an operating voltage between approximately 6.0 V and 8.4 V, a buck converter is required to efficiently reduce the battery voltage to the regulated 3.3 V supply required by some of the other components. Compared to a linear regulator, the switching regulator significantly improves efficiency, reducing power dissipation and extending battery operating life [SSS32].

The input stage consists of a 1 μ F ceramic capacitor connected across the supply rails to reduce input voltage fluctuations and suppress switching noise. The switching frequency of the LM5160 is established by a 68.1 k Ω resistor connected to the RON pin; a resistor divider connected to the EN/UVLO pin establishes the undervoltage lockout threshold. In addition, a 22 nF capacitor connected to the soft-start pin gradually increases the output voltage during startup, reducing stress on the battery and downstream circuitry [SSS33].

A 10 nF bootstrap capacitor connected between the BST and SW pins provides the gate drive voltage required for the internal high-side MOSFET, while a 1 μ F capacitor connected to the VCC pin provides local decoupling for the regulator's internal circuitry. Energy transfer is achieved using a 10 μ H inductor connected to the switching node, with a 100 μ F output capacitor filtering the switching waveform to produce a stable DC output. The large output capacitance was selected to minimize output voltage ripple [SSS33].

The output voltage is regulated through a feedback network consisting of 9.76 k Ω and 1.50 k Ω resistors connected to the FB pin. An 18 nF feed-forward capacitor improves transient response, while a 110 m Ω series resistor provides the ripple signal required for the LM5160's control architecture [SSS33].

6.2 Brake Light and Turn Lights circuits

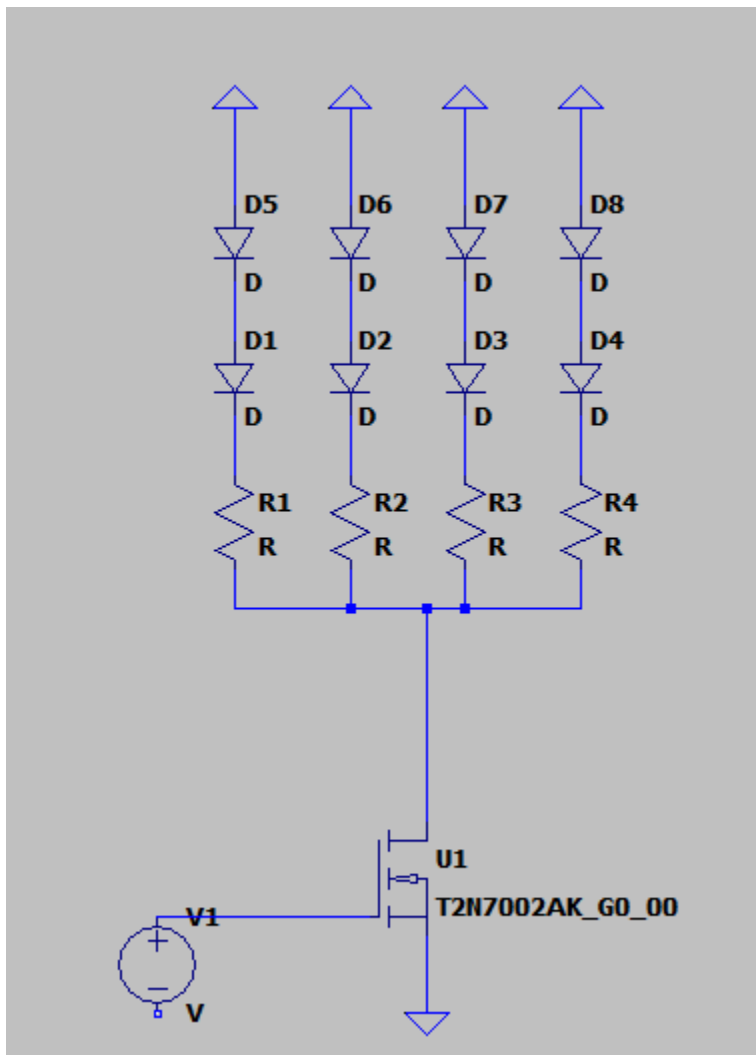


Figure 6.2.1 Brake Light and Turn Lights circuits

The brake light will have 1 circuit with red lights and the turn signal will have 2 copies of this circuit with orange lights.

The brake light will be on its own dedicated pcb along with the left and right turn lights. Both Lights will be comprised of a 4 by 2 array of having 4 columns and 2 rows of L.E.Ds. These L.E.Ds will be red for the brake light and orange for the turn lights. The brake light will be connected to the micron controller which will send a voltage when the brake is pressed. The turn signals will work in a similar way with motorcycle style switch with 3 modes, left, right, and off. When switching to left or right the corresponding brake light will be turned on by a MOSFET controlled by the microcontroller and will blink. To ensure the L.E.Ds do

not get burned a resistor will be sized based off the L.E.D current equation for each L.E.D column.

7.0 Software Design

The BEACON project employs three major, custom-made software components. The most important of the three is the under-seat MCU firmware. The under-seat board serves as the brain for the project, handling the heaviest processing load. The MCU runs a loop architecture, similar to a finite-state machine, with hardware interrupt overrides to ensure any safety threats take priority and will alert the rider. The next software component is the handlebar remote MCU firmware, which is focused on energy conservation and telemetry display. Like the main MCU firmware, the handlebar firmware is primarily in low-power mode and utilizes wait-for-interrupt instructions. Lastly, the display software interprets data from the peripherals and highlights relevant information to the rider.

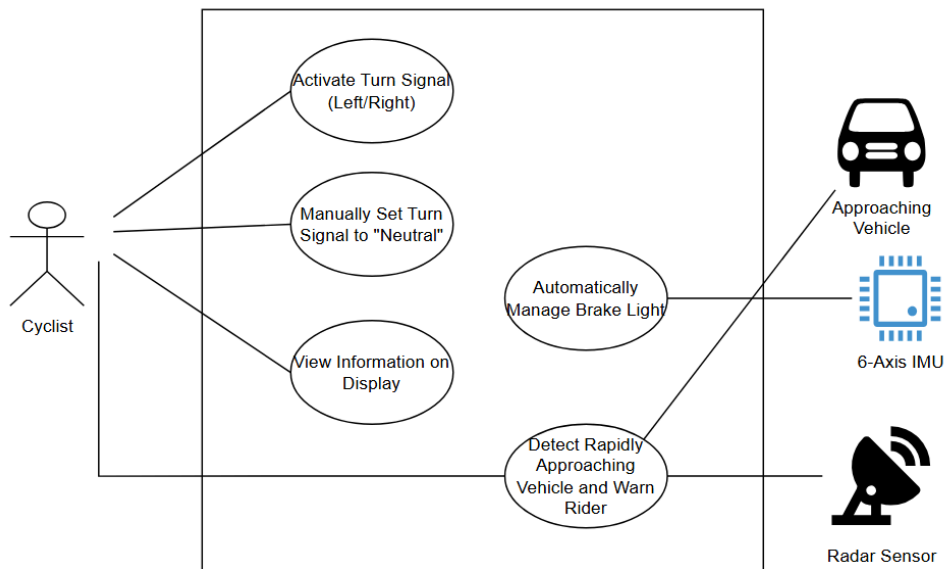


Figure 7.1 Software Use Case Diagram

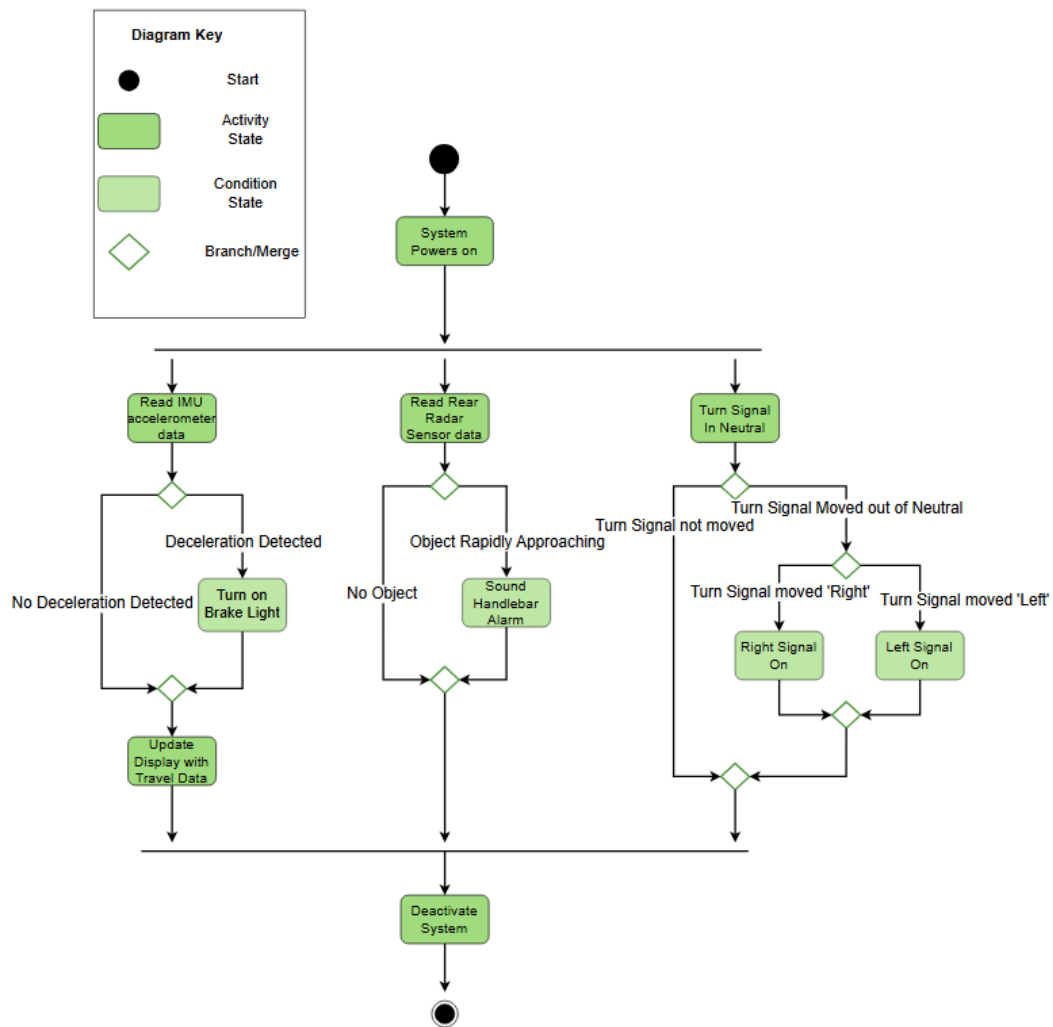


Figure 7.2 Activity Diagram

7.1 Radar Object Detection Algorithm

For the BEACON project, to implement a real-time radar detection safety monitor requires offloading high frequency signal processing to a radar module's dedicated hardware accelerator that is communicating with the underseat MCU. Instead of consuming massive amounts of CPU power to calculate the raw radio frequency transforms, the underseat MCU will receive the already processed data from the Frequency Modulated Continuous Wave (FMCW) radar module. Through the UART communication protocol, the MCU will store and determine the proper procedure based on the binary frames received from the radar module. By utilizing this methodology, the BEACON project will maintain proper response time and will act accordingly to approaching objects detected by the rear radar module.

The processing pipeline being executed on the underseat MCU is divided into individual execution intervals, ideally running within a 50-60 millisecond loop. To prevent the destabilization of time-critical safety features from bus latency, the incoming data bytes will be streamed directly into a circular ring buffer through Direct Memory Access (DMA) [TTT19]. A circular ring buffer, or circular queue, is a fixed-size data structure that uses a single block of memory as if it were connected end-to-end [TTT24]. By implementing this First-In, First-Out (FIFO) data structure, the incoming data bytes being streamed through UART to the MCU will not cause any bottlenecks or other data disruptions. In a danger-detection system, removing all potential causes of data stalls is paramount. From there, the buffer will be scanned for unique data packets through a finite state machine (FSM). The FSM will validate the transmitted packet by verifying packet length declarations through an XOR checksum. Corrupted or partial data packets are instantly dropped, ensuring that the tracking mechanism can only operate on verified targets.

Once a data packet passes the validation threshold, the MCU will load the raw binary into a struct dedicated to tracking data. Thankfully, the radar module does all of the heavy lifting. Therefore, the data that is received by the MCU is the pre-filtered matrix of object data. The MCU will then map these data points from their binary formats into physical variables through the use of two's complement. This will extract the absolute range and relative Doppler velocity for each data point.

7.2 IMU Acceleration Tracking and Adaptive Braking Algorithm

For an automated brake light system for the BEACON project, choosing an effective velocity model means considering response time against filtering of environmental noise. A reliable metric for identifying true braking behavior is the persistent negative acceleration magnitude over a rolling time window. Through the 3-axis accelerometer embedded within the Inertial Measurement Unit (IMU), by detecting a sudden change in linear velocity, the MCU will be able to respond accordingly and engage the rear brake light. Once the MCU receives proper data to confirm that the bike is no longer slowing down, the brake light will be turned off. Through the use of the I2C communication protocol, the MCU and IMU can establish a controller-peripheral relationship. Using I2C fast-mode, the strict execution timing can be met and ensure proper functionality.

The IMU will send linear acceleration telemetry to the underseat MCU, at which point, there are several different approaches that could be implemented algorithmically. For raw signal conditioning and vibration filtering, typical algorithm solutions are moving average filters, which takes the average of a number of points from an input signal to produce each point in the output signal [25], or through an infinite impulse response filter, which allows low-frequency signals to pass while reducing high-frequency noise [TTT25]. Since it is likely that someone using the BEACON project will be on several hour trips, we need to ensure

long-term sensor accuracy. For baseline calibration, static measurements can be taken in different orientations, which is referred to as a tumble test [TTT24]. The tumble test tests the accelerometer by aligning each sensor axis to the direction of, opposite of, and perpendicular to known gravity, providing a measurement of $\pm 1g$ as it rotates [TTT26].

The MCU will overlook the entire communication lifecycle. Targeting the IMU sensor's internal register map through the unique 7-bit hardware address. Once the MCU is connected and pointing to the acceleration data registers, it can issue a repeated start condition, ensuring the transaction is in read mode. To minimize bus overhead and latency, the MCU will read the high and low data bytes for all three dimensions sequentially in a single burst. Once the binary package is transferred over to the MCU's local buffers, the firmware will process the incoming telemetry data. To combine the two, byte sized packets into a singular 2-byte packet, the MCU internal firmware will shift the high byte left by 8 bits, then combine the two byte packets through an OR operation. This will produce signed integer values that can be mapped into physical gravitational units based on the sensor's pre-configured full-scale sensitivity profile. Lastly, the MCU can isolate the longitudinal travel axis, which is typically either the Y-axis or Z-axis depending on the final pcb orientation.

Because bicycles are subjected to constant vibrations from traveling conditions, the MCU does not react to individual raw readings. Instead, the scaled forward acceleration data is fed into a data smoothing algorithm, such as a rolling moving average filter, to suppress high-frequency environmental noise. The MCU continuously evaluates the smoothed baseline against a calibrated negative acceleration threshold (e.g., $-0.15g$). To prevent false positives, such as from brief coasting adjustments, the MCU will implement an algorithm to fuse sequential data points over a moving time frame. Typically, a temporal window integration (TWI) algorithm would be considered. The TWI algorithm will require filtered signals to remain below the calibrated deceleration threshold for a consecutive number of sample cycles. Based on these conditions being met, the MCU can finally classify the event as an intentional braking from the rider.

10.0 Administrative Content

10.1 Budget

The estimated budget for the Beacon project is approximately \$400. This budget is intended to cover the cost of electronic components, printed circuit board (PCB) fabrication, sensors, microcontrollers, batteries, and other hardware required to build and test the prototype, as well as the final design. Cost was considered throughout the component selection process to ensure the final design remains both practical and financially feasible while meeting the project's requirements.

10.2 Bill of Materials

10.2.1 Bill of Materials Table

Component	Part Name	Quantity	Price per Unit
Microcontroller	ESP32-S3-WROOM-1	2	\$6.76
IMU sensor	TBD	1	X
FMCW Sensor	Acconeer XM125	1	\$22
LCD Display	TBD	X	X
Batteries	NMC lithium-ion batteries	X	X
Antenna	Integrated ESP32 Antenna	0	\$0
Signal L.E.D	OSRAM OSOLON SSL 120 Amber	X	\$1.62
Brake L.E.D	TBD	X	X
Turn Signal	TBD	X	X
Test Bicycle	Test Bicycle	1	\$0
PCB 1	TBD	1	TBD
PCB 2	TBD	1	TBD
PCB 3	TBD	1	TBD
Misc.	Miscellaneous parts (wires, resistors, etc.)	Need further Research	Need Further Research

10.3 Distribution of Work

Table X shows the distribution of work among the project team members. Tasks were divided as evenly as possible based on each member's responsibilities and areas of focus to ensure a balanced workload throughout the project. Although individual members were assigned primary responsibility for specific sections, many aspects of the project, including design decisions, research, and report development, were completed collaboratively to ensure consistency and incorporate input from the entire team.

10.3.1 Distribution of Work Tables

Electrical Engineering	Responsibilities
Benjamin Eppl	• Battery Research • Antenna Research • Executive summary • Break light circuit • Turn signal circuit
Sarah McDonald	• Motivation and Background • House of Quality • Prototype Renders • Lighting Research • Acknowledgement of A.I • Standards and Design Constraints • Voltage Regulators
Mia Shoemaker	• MCU, IMU, FMCW Radar sensor research and part selection • Objectives and Goals • Milestones/Budget

Computer Engineering	Responsibilities
Trevor Mongar	• Communication Protocol Research • Software Research • Algorithm Design • Firmware Design

10.4 Milestones

10.4.1 Senior Design 1 Milestones Table

Task	Description	Complete by / Due date
Divide & Conquer Report & Review Committee	Complete 10-page project proposal report along with recruiting	05/29/2026

	review committee members from ECE faculty	
Divide & Conquer Group Meeting	30-minute meeting to discuss project proposal	06/01/2026
30-Page Checkpoint	25% of Final report completed, 30 pages	06/19/2026
60-Page Checkpoint	Halfway point: Project description + Research & Investigation completed	07/03/2026
80-Page Checkpoint	75% of Final report completed, 80 pages	07/06/2026
Final Report	Final SD1 document, at least 120 pages	07/28/2026

10.4.2 Senior Design 2 Milestones Table

Task	Description	Complete by/Due date
Order PCB	Finalize PCB design and order any remaining components	Date TBD, Estimated time 2 weeks
Assemble PCB	Assemble any hardware and PCB	Date TBD, Estimated time 2 weeks
Software integration	Integrate braking detection, turn signals, etc.	Date TBD, Estimated time 2 weeks
Testing and Debugging	Test subsystem functionality, address any hardware/software issues	Date TBD, Estimated time 2 weeks
Prototype/Testing	Fully assemble project prototype and perform outdoor testing	Date TBD, Estimated time 2 weeks

Final Presentation	Complete Final report and Senior design showcase	Date TBD, Estimated time 2 weeks
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8.0 Appendices

[SSS1] National Association of City Transportation Officials, Urban Bikeway Design Guide. New York, NY, USA: NACTO, Jun. 27, 2012. [Online]. Available: <https://nacto.org/publication/urban-bikeway-design-guide/> . [Accessed: Jun. 22, 2026].

[SSS2] World Health Organization, Global Status Report on Road Safety 2023. Geneva, Switzerland: World Health Organization, 2023.

[SSS3] Vision Zero Network, “Vision Zero Network.” [Online]. Available: <https://visionzeronetwork.org> . [Accessed: Jun. 22, 2026].

[SSS4] National Highway Traffic Safety Administration, “Bicycle Safety,” Dec. 18, 2018. [Online]. Available: <https://www.nhtsa.gov/road-safety/bicycle-safety> . [Accessed: Jun. 22, 2026].

[SSS5] A. N. Jannah et al., “Cyclist safety in the digital age: A review of advanced warning technologies,” Accident Analysis & Prevention, vol. 217, p. 108069, 2025, doi: 10.1016/j.aap.2025.108069.

[SSS6] IBM, “What are large language models (LLMs)?,” IBM Think. [Online]. Available: <https://www.ibm.com/think/topics/large-language-models>. [Accessed: Jun. 22, 2026].

[SSS7] C. Zhai, S. Wibowo, and L. D. Li, “The effects of over-reliance on AI dialogue systems on students’ cognitive abilities: A systematic review,” Smart Learning Environments, vol. 11, no. 1, art. 28, 2024. doi: 10.1186/s40561-024-00316-7. Available: <https://doi.org/10.1186/s40561-024-00316-7>. [Accessed: Jun. 22, 2026].

[SSS8] National Highway Traffic Safety Administration, “Bicycle Safety,” NHTSA. [Online]. Available: <https://www.nhtsa.gov/road-safety/bicycle-safety>. [Accessed: Jun. 22, 2026].

[SSS9] Washington State Legislature, “RCW 46.61.780 — Lamps and other equipment on bicycles,” Revised Code of Washington. [Online]. Available: <https://app.leg.wa.gov/rcw/default.aspx?cite=46.61.780>. [Accessed: Jun. 22, 2026].

[SSS10] Laser Scientist, “Electroluminescence.” [Online]. Available: <https://laserscientist.com/electroluminescence/>. [Accessed: Jun. 22, 2026].

[SSS11] Kingbright USA, “WP7113ID LED Datasheet.” [Online]. Available: <https://www.kingbrightusa.com/images/catalog/SPEC/WP7113ID.pdf>. [Accessed: Jun. 22, 2026].

[SSS12] Lumileds, “High Power LEDs.” [Online]. Available: <https://lumileds.com/products/high-power-leds/>. [Accessed: Jun. 22, 2026].

[SSS13] Vishay Semiconductor, “TSAL6200 Infrared Emitting Diode Datasheet.” [Online]. Available: <https://www.vishay.com/docs/81010/tsal6200.pdf>. [Accessed: Jun. 22, 2026].

[SSS14] U.S. Department of Energy, “LED Basics,” Energy.gov. [Online]. Available: <https://www.energy.gov/cmei/ssl/led-basics>. [Accessed: Jun. 22, 2026].

[SSS15] Lamp Insider, “How Much Does LED Light Cost?” [Online]. Available: <https://lampinsider.com/how-much-does-led-light-cost/>. [Accessed: Jun. 22, 2026].

[SSS16] Lamp Insider, “How Does an Incandescent Lamp Work?” [Online]. Available: <https://lampinsider.com/how-does-an-incandescent-lamp-work/>. [Accessed: Jun. 22, 2026].

[SSS17] Quick and Easy Lighting, “Lumens Chart.” [Online]. Available: <https://quickandeasylighting.com/lumens-chart/>. [Accessed: Jun. 22, 2026].

[SSS18] arXiv, “arXiv:2010.10584.” [Online]. Available: <https://arxiv.org/pdf/2010.10584>. [Accessed: Jun. 22, 2026].

[SSS19] Headlight Bulb Kit, “How Long Do Incandescent Light Bulbs Last?” [Online]. Available: <https://headlightbulbkit.com/how-long-do-incandescent-light-bulbs-last/>. [Accessed: Jun. 22, 2026].

[SSS20] Electrical4U, “Fluorescent Lamp: Its Working Principle.” [Online]. Available: <https://www.electrical4u.com/fluorescent-lamp-its-working-principle/>. [Accessed: Jun. 22, 2026].

[SSS21] Cornell Law School Legal Information Institute, “10 CFR § 430.32.” [Online]. Available: <https://www.law.cornell.edu/cfr/text/10/430.32>. [Accessed: Jun. 22, 2026].

[SSS22] Bright Light Hub, “How Long Do Fluorescent Bulbs Last?” [Online]. Available: <https://brightlighthub.com/how-long-do-fluorescent-bulbs-last/>. [Accessed: Jun. 22, 2026].

[SSS23] Lumileds, “Product Image / Adaptive Media Asset (73c0bbb3dad3f9c1782bdc33f0d654e8ce3a507e).” [Online]. Available: <https://otmm.lumileds.com/adaptive-media/73c0bbb3dad3f9c1782bdc33f0d654e8ce3a507e>. [Accessed: Jun. 22, 2026].

[SSS24] Cree LED, "XLamp XPE2 LED Datasheet." [Online]. Available: <https://downloads.cree-led.com/files/ds/x/XLamp-XPE2.pdf>. [Accessed: Jun. 22, 2026].

[SSS25] ams OSRAM, "OSLON Signal Product Family." [Online]. Available: <https://ams-osram.com/products/product-families/oslon-signal>. [Accessed: Jun. 22, 2026].

[SSS26] International Organization for Standardization, Cycles—Lighting and Retro-Reflective Devices—Part 1: Lighting and Light Signalling Devices, ISO 6742-1:2023, Geneva, Switzerland, 2023.

[SSS27] J. D. Wood, P. Lacherez, and J. Marszalek, "Drivers' and cyclists' experiences of sharing the road: Incidents, attitudes and perceptions of visibility," *Accident Analysis & Prevention*, vol. 78, pp. 88–96, 2015.

[SSS28] International Electrotechnical Commission, *IEC 60529: Degrees of Protection Provided by Enclosures (IP Code)*, 2nd ed., Geneva, Switzerland: IEC, 1989+A1:1999+A2:2013.

[SSS29] H. Katz, "Environmental Protection of Electronic Equipment: Understanding the IP Code," *IEEE Industry Applications Magazine*, vol. 20, no. 2, pp. 32–39, Mar./Apr. 2014.

[SSS30] IPC, "IPC-2221B: Generic Standard on Printed Board Design", Bannockburn, IL, USA, 2012.

[SSS31] C. F. Coombs Jr., "Printed Circuits Handbook", 7th ed. New York, NY, USA: McGraw-Hill, 2016.

[SSS32] Texas Instruments, "WEBENCH® Power Designer - Switching Regulator Designer," Texas Instruments, Dallas, TX, USA. [Online]. Available: <https://webench.ti.com/power-designer/switching-regulator/select>. [Accessed: Jul. 5, 2026].

[SSS33] Texas Instruments, "LM5160, LM5160A Wide Input 65-V, 2-A Synchronous Buck / Fly-Buck™ DC/DC Converter," Rev. E, Oct. 2018. [Online]. Available: <https://www.ti.com/product/LM5160>. [Accessed: Jul. 5, 2026].

[MMM1] Garmin, “Bike Radar and Tail Light – Varia™ RTL515.” Accessed: May 28, 2026. [Online]. Available: <https://www.garmin.com/en-US/p/698001/>

[MMM2] UNIT 1, “Best Bike Lights with Turn Signals & Brake Lights for E-Bikes.” Accessed: May 28, 2026. [Online]. Available: <https://www.unit1gear.com/pages/smart-light>

[MMM3] CYCL, “WingLights Bicycle Turn Signals.” Accessed: May 28, 2026. [Online]. Available: <https://cycl.bike/collections/winglights>

[MMM4] WatElectronics, “What Is Ultrasonic Sensor: Working & Its Applications.” Accessed: May 28, 2026. [Online]. Available: <https://www.watelectronics.com/ultrasonic-sensor/>

[MMM5] Matterport, “What Is LiDAR & What Is It Used For? 9 Key Applications.” Accessed: May 28, 2026. [Online]. Available: <https://matterport.com/blog/what-is-lidar>

[MMM6] GeeksforGeeks, “Continuous Wave (CW) Radar.” Accessed: May 28, 2026. [Online]. Available: <https://www.geeksforgeeks.org/electrical-engineering/continuous-wave-radar/>

[MMM7] GeeksforGeeks, “FMCW Radar.” Accessed: May 28, 2026. [Online]. Available: <https://www.geeksforgeeks.org/electronics-engineering/fmcwr-radar/>

[MMM8] Infineon Technologies, *BGT60TR13C: 60 GHz Radar Sensor Datasheet*, Rev. 2.49, Nov. 21, 2023. Accessed: May 28, 2026. [Online]. Available: <https://www.infineon.com/assets/row/public/documents/24/49/infineon-bgt60tr13c-datasheet-en.pdf>

[MMM9] Acconeer AB, *XM125 Entry+ Module Datasheet*, Version 1.0, Mar. 30, 2023. Accessed: May 28, 2026. [Online]. Available: https://www.mouser.com/datasheet/2/1126/XM125_datasheet-3133248.pdf

[MMM10] Seeed Studio, *MR60BHA2 60 GHz mmWave Human Detection Sensor Datasheet*. Accessed: May 28, 2026. [Online]. Available: <https://www.mouser.com/datasheet/3/6064/1/3343423531423037464441323245303237413231453538434335384243414438.pdf>

[MMM11] GeeksforGeeks, “Microcontroller and Its Types.” Accessed: May 28, 2026. [Online]. Available: <https://www.geeksforgeeks.org/digital-logic/microcontroller-and-its-types/>

[MMM12] IBM, “What Are Field-Programmable Gate Arrays (FPGAs)?” Accessed: May 28, 2026. [Online]. Available: <https://www.ibm.com/think/topics/field-programmable-gate-arrays>

[MMM13] Science Insights, “What Is a Single-Board Computer and What It’s Used For.” Accessed: May 28, 2026. [Online]. Available: <https://scienceinsights.org/what-is-a-single-board-computer-and-what-its-used-for/>

[MMM14] Espressif Systems, *ESP32-S3-WROOM-1/ESP32-S3-WROOM-1U Datasheet*, Version 1.7, May 10, 2024. Accessed: May 28, 2026. [Online]. Available: https://documentation.espressif.com/esp32-s3-wroom-1_wroom-1u_datasheet_en.pdf

[MMM15] Texas Instruments, *MSP430FR698x(1), MSP430FR598x(1) Mixed-Signal Microcontrollers Datasheet*, Rev. D, Aug. 2018. Accessed: May 28, 2026. [Online]. Available: <https://www.ti.com/lit/ds/symlink/msp430fr6989.pdf>

[MMM16] STMicroelectronics, *STM32F446xC/E ARM® Cortex®-M4 32-bit MCU+FPU, 225 DMIPS, up to 512 Kbytes Flash/128 Kbytes RAM Datasheet*, Rev. 7, Jan. 2021. Accessed: May 28, 2026. [Online]. Available: <https://www.st.com/resource/en/datasheet/stm32f446re.pdf>

[BBB1] Bluetooth SIG, "Part A: Physical Layer Specification," Bluetooth Core Specification, Low Energy Controller Physical Layer Specification. [Online]. Available:

<https://www.bluetooth.com/wp-content/uploads/Files/Specification/HTML/core-54/out/en/low-energy-controller/physical-layer-specification.html>. [Accessed: 2-Jul-2026].

[BBB2] Espressif Systems, "ESP32-S3-WROOM-1 & ESP32-S3-WROOM-1U Datasheet," Espressif Systems. [Online]. Available: https://documentation.espressif.com/esp32-s3-wroom-1_wroom-1u_datasheet_en.pdf. [Accessed: 25-June-2026].

[BBB3] Espressif Systems, "PCB Layout Design - ESP32-S3," ESP Hardware Design Guidelines, Espressif Systems. [Online]. Available: <https://docs.espressif.com/projects/esp-hardware-design-guidelines/en/latest/esp32s3/pcb-layout-design.html>. [Accessed: 28-June-2026].

[BBB4] Taoglas, "FXP73 Blue Diamond 2.4GHz Flexible PCB Antenna," datasheet, part no. FXP73.07.0100A. [Online]. Available: <https://www.taoglas.com/datasheets/FXP73.07.0100A.pdf>. [Accessed: 3-Jul-2026].

[BBB5] Johanson Technology, "Chip Antenna Booklet for RF Engineers," Johanson Technology. Accessed: Jul. 1, 2026. [Online]. Available: <https://www.johansontechnology.com/tech-notes/understanding-chip-antennas/>

[BBB6] Taoglas, "Freedom FXP72 2.4GHz Flex PCB Antenna, 53mm Ø1.13, I-PEX MHF® I (U.FL)," Taoglas. Accessed: Jul. 1, 2026. [Online]. Available: <https://www.taoglas.com/product/freedom-fxp72-2-4ghz-flex-pcb-antenna/>

[BBB11] Battery University, "BU-205: Types of Lithium-ion," Battery University, Dec. 8, 2023. [Online]. Available: <https://www.batteryuniversity.com/article/bu-205-types-of-lithium-ion>. [Accessed: June. 19, 2026].

[BBB12] Battery University, "BU-701: How to Prime Batteries," Battery University, Oct. 27, 2021. [Online]. Available: <https://www.batteryuniversity.com/article/bu-701-how-to-prime-batteries>.

[Accessed: June. 19, 2026].

[BBB13] <https://www.orbtronic.com/content/samsung-35e-datasheet-inr18650-35e.pdf>

[BBB8] Duracell, "DX1500B4N Nickel-Metal Hydride Battery Specifications." [Online]. Available: <https://www.farnell.com/datasheets/1860438.pdf>.

[BBB9] Elsevier B.V., "Nickel-Metal Hydride Battery," ScienceDirect Topics. [Online]. Available: <https://www.sciencedirect.com/topics/engineering/nickel-metal-hydride-battery>.

[BBB13] Samsung SDI, "INR18650-35E Datasheet," *Samsung SDI*. [Online]. Available: <https://www.orbtronic.com/content/samsung-35e-datasheet-inr18650-35e.pdf>. [Accessed: June. 20, 2026].

[BBB14] LiPol Battery Co., Ltd., "Lithium Polymer Battery 3.7V 30–10000mAh," LiPol Battery Co., Ltd. [Online]. Available: <https://www.lipolbattery.com/lithium%20polymer%20battery.html>. [Accessed: 4-Jul-2026].

[BBB15] **Prompt:** please give me sources to look into myself for lithium batteries for my senior design project beacon
Response:

For a Senior Design report, your professors will generally prefer **manufacturer data sheets, university research centers, IEEE papers, and technical battery references** over blogs or shopping websites.

Here are the best sources to start with:

1. Battery University (Industry Standard Reference)

This is one of the most commonly cited battery education resources in engineering.

Types of Lithium-Ion Batteries

- [Battery University - Types of Lithium-Ion Batteries](#)
- Covers:
 - Li-ion (NMC, NCA, LCO)
 - LiFePO₄
 - Energy density
 - Safety
 - Cycle life

How Lithium Batteries Work

- [Battery University - How Lithium Batteries Work](#)

Why Lithium-Ion Became Popular

- [Battery University - Is Lithium-Ion the Ideal Battery?](#)

2. University of Washington Clean Energy Institute

Excellent academic source.

- [UW Clean Energy Institute - Lithium-Ion Battery Overview](#)

Useful facts:

- Li-ion batteries have no memory effect.
- Typical cell voltage is about 3.6 V.
- Higher energy density than many competing technologies.

3. Panasonic 18650 Cell Documentation

Since you're considering a 7.4 V battery pack made from 18650 cells, looking at real commercial cells is valuable.

- [Panasonic Official Site](#)

Search for:

- NCR18650B
- NCR18650GA

These are some of the most widely used Li-ion cells and provide:

- Capacity
- Weight
- Voltage curves
- Safety limits

4. Compare Li-Ion vs LiPo

Since your project could theoretically use either technology, you'll want a comparison source.

- [LiPo vs Li-Ion Comparison](#)
- [Battery Power Tips - Li-Ion vs LiPo FAQ](#)

Key takeaway:

- LiPo batteries are lighter and can be custom-shaped.
- Li-ion cylindrical cells generally provide better durability, safety, and lower cost for a bicycle electronics project.

5. Compare Li-Ion vs LiFePO4

This is probably the most important comparison for your report.

- [NiMH vs Li-Ion vs LiFePO4 Comparison](#)

Research:

- Energy density

- Cycle life
- Safety
- Weight

Typical conclusion:

- LiFePO4 is safer and lasts longer.
- Li-ion provides higher energy density and lower weight, making it a better fit for compact bicycle electronics.

IEEE Search Terms

Use these exact searches in the [IEEE Xplore Digital Library](#):

- "Lithium-ion battery energy density comparison"
- "18650 lithium-ion battery performance"
- "Lithium-ion battery management systems"
- "LiFePO4 versus lithium-ion batteries"
- "Battery selection for embedded systems"
- "Rechargeable battery technologies for portable electronics"

[BBB16] **Prompt:** For my senior design project BEACON I now need to research on the antenna used for this project and go through specific ones I could use for the project

Response: Based on your report, the antenna section should focus on the **BLE wireless link between the handlebar PCB and the under-seat PCB**, not the radar antenna. Your system already lists **two PCBs**, one under the seat and one on the handlebar, with a **BLE 5.0 wireless module** and a required **wireless communication range of at least 10 ft**.

Best direction for B.E.A.C.O.N.

For your project, I would recommend using a **2.4 GHz BLE antenna**, since Bluetooth Low Energy operates in the **2.4 GHz ISM band from 2400 MHz to 2483.5 MHz** and uses 40 RF channels. Because the handlebar-to-seat distance is very short, your project does **not** need a large external antenna. The best practical choice is either:

Best prototype choice: a BLE module with a built-in PCB antenna

Best final custom PCB choice: a 2.4 GHz PCB trace antenna or a compact chip antenna

Best fallback if enclosure placement hurts range: a small FPC antenna with U.FL/I-PEX cable

3.X Antenna Selection

The B.E.A.C.O.N system requires wireless communication between the handlebar-mounted subsystem and the under-seat main electronics module. Since the system specification identifies Bluetooth Low Energy 5.0 as the intended wireless communication method, the antenna used for this portion of the design must operate within the Bluetooth Low Energy frequency range. Bluetooth Low Energy operates in the 2.4 GHz ISM band from 2400 MHz to 2483.5 MHz and uses 40 RF channels spaced 2 MHz apart. Because the required communication distance for B.E.A.C.O.N is only 10 ft, the antenna does not need to support long-range communication. Instead, the antenna should prioritize compact size, low cost, low power operation, and reliable communication between the two bicycle-mounted modules.

Several antenna types were considered for the B.E.A.C.O.N wireless subsystem, including integrated PCB antennas, PCB trace antennas, ceramic chip antennas, and flexible PCB antennas. The simplest option is to use a BLE module with an integrated antenna. Modules such as the ESP32-C3-MINI-1 include a built-in PCB antenna, while similar variants such as the ESP32-C3-MINI-1U provide an external antenna connector for applications requiring a different antenna placement. This approach reduces antenna design complexity because the RF layout is already implemented by the module manufacturer. For a senior design prototype, this is a major advantage because improper antenna layout can significantly reduce wireless range and reliability.

Another option is a printed PCB inverted-F antenna. A PCB antenna is attractive because it adds little to no component cost and can be fabricated directly as part of the custom PCB. Texas Instruments provides a 2.4 GHz PCB antenna reference design that occupies approximately 15.2 mm by 5.7 mm of board space and is designed for operation across the 2.4 GHz ISM band. This type of antenna would be useful if the wireless radio is implemented directly on the B.E.A.C.O.N PCB rather than through a pre-certified module. However, PCB antennas require careful layout, a controlled 50 Ω feed line, ground-plane clearance, and proper placement at the edge of the board. If these layout requirements are not followed, the antenna efficiency and wireless range may be reduced.

Ceramic chip antennas were also considered because they provide a compact surface-mount solution. The Johanson 2450AT18B0100001E is an example of a 2.4 GHz chip antenna intended for Bluetooth, WLAN, and similar ISM-band applications. This type of antenna is useful when PCB space is limited because it is much smaller than most printed trace antennas. However, chip antennas still require a matching network and a layout that follows the manufacturer's recommended ground clearance. As a result, they are not completely plug-and-play and may require tuning or testing to achieve strong RF performance.

Flexible PCB antennas were also evaluated as an alternative for cases where the antenna needs to be placed away from the main PCB. The Taoglas

FXP73.07.0100A is a 2.4 GHz flexible PCB antenna designed for Wi-Fi, ZigBee, Bluetooth, and ISM-band applications. Its thin flexible structure allows it to be mounted inside a plastic enclosure, away from batteries, metal bicycle parts, or dense PCB components that could interfere with antenna radiation. This makes it a strong backup option if the under-seat enclosure or bicycle frame reduces wireless performance during testing. The disadvantage is that it requires a cable and U.FL/I-PEX connector, which increases cost, assembly complexity, and required enclosure space.

For the B.E.A.C.O.N project, the recommended antenna solution is to use a BLE module with an integrated PCB antenna for the first prototype. This choice best matches the project requirements because the required range is short, the design must remain compact, and reducing RF design risk is important. If range testing shows that the under-seat enclosure, battery pack, or bicycle frame interferes with communication, then a U.FL-based flexible PCB antenna such as the Taoglas FXP73 or Molex 146153 series can be used as a secondary option. This approach allows the project to begin with the simplest and lowest-risk antenna design while still leaving a reliable upgrade path if real-world testing requires improved antenna placement.

My recommendation

Use a **BLE module with a built-in PCB antenna** first. It is the safest choice for senior design because your range requirement is only **10 ft**, and it avoids needing to tune RF hardware. For the report, compare it against the **TI PCB inverted-F antenna**, **Johanson chip antenna**, and **Taoglas/Molex flexible antennas**, then conclude that the built-in antenna module is the best prototype choice.

[BBB16] Sanyo Energy, "Specifications for NCR18650GA," Panasonic/Sanyo, lithium-ion rechargeable battery datasheet. [Online]. Available: <https://www.orbtronic.com/content/Datasheet-specs-Sanyo-Panasonic-NCR18650GA-3500mah.pdf>. [Accessed: 27-June-2026].

[BBB17] LG Chem, "Lithium Ion INR18650 MJ1 3500mAh Cell Specification," LG Chem/Tenergy, rechargeable lithium-ion battery datasheet. [Online]. Available: https://power.tenergy.com/content/datasheet/30708_datasheet.pdf. [Accessed: 14-June-2026].

[BBB18] Taoglas, "FXP72.07.0053A: 2.4GHz Freedom Flexible PCB Antenna with 53mm 1.13 Cable and IPEX MHFI (U.FL) Connector," Datasheet, Rev. G, Feb. 28, 2025. [Online]. Available: <https://www.taoglas.com/datasheets/FXP72.07.0053A.pdf>. [Accessed: 5-Jul-2026].

[BBB19] Johanson Technology, “2440AT62B0085002U: 2.44 GHz RF Chip Antenna, RHCP, AEC-Q200,” Datasheet, Rev. 2.0, 2025. [Online]. Available: <https://www.johansontechnology.com/docs/4405/Antenna-2440AT62B0085002U-Rev2.0.pdf>. [Accessed: 5-Jul-2026].

[TTT1] E. Pena and M. G. Legaspi, “UART: A Hardware Communication Protocol Understanding Universal Asynchronous Receiver/Transmitter | Analog Devices,” Analog.com, 2024. <https://www.analog.com/en/resources/analog-dialogue/articles/uart-a-hardware-communication-protocol.html>. [Accessed: 9-Jun-2026].

[TTT2] “Universal Asynchronous Receiver Transmitter (UART)”. Texas Instruments. https://dev.ti.com/tirex/explore/node?isTheia=false&node=A_Afl-q.bte68MLrOgncnKrw_C29X-ACADEMY_KgQzuuf_LATEST (accessed June 6, 2026)

[TTT3] J. Wu, “A Basic Guide to I2C.” Accessed: Jun. 11, 2026. [Online]. Available: https://www.ti.com/lit/an/sbaa565/sbaa565.pdf?ts=1781071511328&ref_url=https%253A%252F%252Fwww.google.com%252F

[TTT4] SFUPTOWNMAKER, “I2C - learn.sparkfun.com,” Sparkfun.com, 2018. <https://learn.sparkfun.com/tutorials/i2c/all>

[TTT5] STMicroelectronics, “6-axis IMU: always-on 3-axis accelerometer and 3-axis gyroscope with ISPU - intelligent sensor processing unit”, ISM330IS datasheet, Jun. 2022 [Revised Aug. 2023].

[TTT6] Di: Manuel Jiménez, R. Palomera, Isidoro Couvertier, and Springerlink (Online Service, Introduction to Embedded Systems : Using Microcontrollers and the MSP430. Pubblicazione: New York, Ny: Springer New York, 2014.

[TTT7] “What is SPI ? : The Basics of Serial Peripheral Interface - Keysight Oscilloscope Glossary - Keysight Technologies,” Keysight.com, 2026. <https://www.keysight.com/used/us/en/knowledge/glossary/oscilloscopes/what-is-spi-the-basics-of-serial-peripheral-interface>

[TTT8] Cranes Varsity, “Why is Embedded C Programming Essential for Every Embedded Systems Engineer in 2025? - Cranes Varsity,” Cranes Varsity, Jul. 21, 2025. <https://cranesvarsity.com/why-is-embedded-c-programming-essential-for-every-embedded-systems-engineer-in-2025/>

- [TTT9] M. Barr and A. Massa, Programming Embedded Systems: With C and GNU Development Tools, 2nd ed. Sebastopol, CA, USA: O'Reilly Media, 2006.
- [TTT10] Information Technology — Programming Languages, Their Environments and System Software Interfaces — Extensions for the Programming Language C to Support Embedded Processors, ISO/IEC JTC1/SC22 N35 74, PDTR 18037, 2002. [Online]. Available: <https://www.open-std.org/jtc1/sc22/open/n3574.pdf>
- [TTT11] P. A. Laplante and S. J. Ovaska, Real-Time Systems Design and Analysis: Tools for the Practitioner, 4th ed. Hoboken, NJ, USA: Wiley-IEEE Press, 2012. [Online]. Available: <https://staff.emu.edu.tr/alexanderchefranov/Documents/CMSE443/CMSE443%20Spring2020/Laplante2012%20Real-Time%20Systems%20Design%20and%20Analysis.pdf>
- [TTT12] P. P. Devkota and R. Chitrakar, "Dynamic Memory Allocation: Implementation and Misuse," Medicon Engineering Themes, vol. 6, no. 4, pp. 29-38, Apr. 2024. [Online]. Available: <https://themedicon.com/pdf/engineeringthemes/MCET-06-204.pdf>
- [TTT13] M. Barr, Embedded C Coding Standard. Germantown, MD, USA: Barr Group, 2018. [Online]. Available: https://barrgroup.com/sites/default/files/barr_c_coding_standard_2018.pdf
- [TTT14] M.F.Juhász, "Modern C++ in Embedded Systems: Utilization in Modern C++ Language Features for Creating Zero-Overhead Firmware Architectures," M.S. thesis, Dept. Comput. Eng., TU Wien, Vienna, Austria, 2023. [Online]. Available: <https://repositum.tuwien.at/bitstream/20.500.12708/190828/1/Juhasz%20Marcell%20Ferenc%20-%202023%20-%20Modern%20C%20in%20embedded%20systems%20-%20Utilization%20of...pdf>
- [TTT15] "Zero-Cost Abstraction in C++ for Embedded Systems," Jbinternational.co.uk, 2016. <https://www.jbinternational.co.uk/article/view/4671> (accessed Jun. 26, 2026).
- [TTT16] A. Sharman, S. Sharma, S. R. Tanksalkar, S. Torres-Arias, and A. Machiry, "Rust for Embedded Systems: Current State and Open Problems," Proceedings of the 2024 on ACM SIGSAC Conference on Computer and Communications Security, pp. 2296-2310, Dec, 2024, doi: <https://doi.org/10.1145/3658644.3690275>.
- [TTT17] "The Rust Programming Language - The Rust Programming Language," doc.rust-lang.org. <https://doc.rust-lang.org/book/title-page.html>

[TTT18] Rust Core Team, "What we heard about Rust's challenges," The Official Rust Programming Language Blog, Mar. 2026. [Online]. Available: <https://blog.rust-lang.org/2026/03/20/rust-challenges/>

[TTT19] G. L. Charvat, Small and Short-Range Radar Systems. Boca Raton, FL USA: CRC Press, 2014.

[TTT20] S. W. Smith, "Moving Average Filters," in The Scientist and Engineer's Guide to Digital Signal Processing, San Diego, CA, USA: California Technical Publishing, 1997, ch. 15, pp. 277–284. [Online]. Available: https://www.analog.com/media/en/technical-documentation/dsp-book/dsp_book_ch15.pdf. [Accessed: Jul. 1, 2026].

[TTT21] S. W. Smith, "Moving Average Filters," in The Scientist and Engineer's Guide to Digital Signal Processing, San Diego, CA, USA: California Technical Publishing, 1997, ch. 19, pp. 319–332. [Online]. Available: https://www.analog.com/media/en/technical-documentation/dsp-book/dsp_book_Ch19.pdf. [Accessed: Jul. 3, 2026].

[TTT22] O. Rioul and M. Vetterli, "Wavelets and signal processing," IEEE Signal Processing Magazine, vol. 8, no. 4, pp. 14–38, Oct. 1991.

[TTT23] W. H. Press, S. A. Teukolsky, W. T. Vetterling, and B. P. Flannery, Numerical Recipes: The Art of Scientific Computing, 3rd ed. Cambridge, UK: Cambridge University Press, 2007, ch. 13.

[TTT24] Siddharth, "Implementing Circular Buffer in C," EmbedJournal, May 2014. [Online]. Available: <https://embedjournal.com/implementing-circular-buffer-embedded-c/#what-is-a-circular-buffer>

[TTT25] O. Hinton, "Design of IIR Filters," in EEE305/EEE801 Part A: Digital Signal Processing, Chapter 5, University of Newcastle upon Tyne, Newcastle, UK, pp. 5.1–5.10. [Online]. Available: <https://www.staff.ncl.ac.uk/oliver.hinton/eee305/Chapter5.pdf>. [Accessed: Jul. 1, 2026].

[TTT26] VectorNav Technologies, "IMU Calibration & Specifications," in Inertial Navigation Primer, VectorNav. [Online]. Available: <https://www.vectornav.com/resources/inertial-navigation-primer/specifications--and--error-budgets/specs-imucal>. [Accessed: Jul. 1, 2026].